Local Networks and Technology Adoption during Industry-Wide Technological Change

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Abstract

Firms' adoption responses to network activity change as the broader industry develops. Using plausibly exogenous peer drilling during the U.S. oil and gas hydraulic-fracturing transition, I compare markets at the same stage of local development. Adoption responses are negative when the technology is young industry-wide but positive in later-developing markets. In those later markets, responses weaken as local experience accumulates. Early investment responses are positive even when adoption responses are negative. A calibrated model reproduces these patterns through adoption costs that decline with industry-wide and local development and diminishing returns to local information.

JEL Classification: O33, G31, D83, D85, L71.

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Firms placed near one another observe each other's choices and their results, transact repeatedly, and share workers who carry knowledge from one operation to the next. A large literature studies these local networks as an important source of productivity growth and technological change.[1] The information these networks provide enters firms' decisions about which technology to use and how much capital to commit to it. Yet a local network rarely grows in isolation. It forms while the broader industry is still learning the technology, adoption is spreading across markets, and performance is improving. The broader transition can change what local information implies about the returns to the technology and how costly it is for firms to act on that information. This paper provides novel evidence that firms' adoption and investment responses change over this transition and uses the contrast between those responses to examine the roles of returns and adoption costs.

New technologies contribute to growth only as firms adopt them, yet adoption is uneven.[2] Some technologies sweep an industry while others stall or never spread. Research on innovation and creative destruction explains how technologies emerge and displace existing production methods. Diffusion also depends on whether firms incorporate those technologies into their own operations. For firms, adoption means committing capital to a change in production technology. That decision is distinct from the amount a firm invests after adopting. Even when information from peers raises the expected return to a technology, acting on it may require fixed expenditures, specialized inputs and organizational changes. Industry-wide development and accumulated local experience can change expected returns and implementation costs at different rates. Firms' responses to local information depend on

[1] Knowledge flows are geographically localized (Jaffe, Trajtenberg and Henderson, 1993; Audretsch and Feldman, 1996), they raise the productivity of nearby plants (Moretti, 2004; Greenstone, Hornbeck and Moretti, 2010), and knowledge sharing is part of why firms cluster in the first place (Ellison, Glaeser and Kerr, 2010). Public R&D also produces geographically concentrated spillovers (Myers and Lanahan, 2022).

[2] For broader evidence on technology adoption and diffusion, see Hall and Khan (2003), Comin and Mestieri (2014) and Kalyani et al. (2021).

the balance between these benefits and costs. A local signal can make additional investment more attractive to firms already using the technology without inducing other firms to adopt. Observing adoption and investment separately makes this tension visible.

I study this question in the U.S. oil and gas hydraulic-fracturing ("fracking") revolution, where a consequential technology spread gradually through incumbent firms as well as entry by new firms. Horizontal drilling and hydraulic fracturing made it possible to extract tight oil from shale and sandstone formations at large scale, with effects on global energy markets.[3] In early 2010, the new technology accounted for about 10% of U.S. oil output. By 2023, almost 70% of output came from fracked wells. Over the same period, U.S. crude oil output rose from 5.5 million barrels per day in 2010 to 12.9 million in 2023.[4]

The transition occurred in waves across oil-producing regions as new formations were discovered. This staggered development makes it possible to compare the effect of a firm's local network across stages of the industry-wide transition. Drilling and permitting records identify when and where firms invest and the technology used in each project. Mandatory production reports track monthly physical output.[5] Together, these data distinguish a firm's choice of production technology from the amount it invests in that technology. I construct firms' nationwide portfolios of technology choices and measure adoption as a shift toward horizontal drilling relative to the firm's previous technology mix. Investment measures whether and how much horizontal drilling occurs. I also use earlier wells to construct measures of the information a local network carries about productivity and implementation.

[3]See Kilian (2017).

[4]The historical hydraulically fractured share is from EIA, "Hydraulic fracturing accounts for about half of current U.S. crude oil production" (March 15, 2016), `https://www.eia.gov/todayinenergy/detail.php?id=25372`, which is based on EIA, IHS Global Insight, and DrillingInfo well-completion and production data. The post-2015 extension uses EIA's Short-Term Energy Outlook Data Browser, Table 10b, tight-oil production by play, `https://www.eia.gov/outlooks/steo/data/browser/`, aggregating tight-oil plays and dividing by total U.S. crude oil production. The total crude-oil output figures are annual averages from EIA Petroleum and Other Liquids, "U.S. Field Production of Crude Oil," series `MCRFPUS2`, annual frequency, 5.48 million barrels per day in 2010 and 12.93 million barrels per day in 2023.

[5]Physical output is rarely observable in firm data. Productivity measures in the literature are typically backed out of sales, so prices and demand movements confound physical efficiency (Foster, Haltiwanger and Syverson, 2008). Here productivity is measured directly from quantities produced.

I combine these measures with plausibly exogenous variation in peer investment to estimate how adoption and investment responses vary with the strength of the local network signal.

The empirical strategy combines local network signals constructed from previously completed development blocks with contractually timed variation in peer activity. I estimate spatial models formation by formation on consecutive 2,000-well blocks and classify each local peer environment using four signals from earlier wells. These signals capture local performance clustering ("peer performance"), the joint explanatory power of peers' observed techniques ("technique intensity"), the number of informative technique attributes ("peer-technique breadth"), and the skill of nearby operators ("peer skill"). For each firm-quarter observation, I use the network measures estimated from the most recently completed block. The variation in peer activity comes from lease expirations. An oil and gas lease typically lasts three to five years and the right to drill expires unless a well is drilled, so expirations concentrate peer drilling at dates set years earlier.[6] I use a bunching estimator, motivated by the bunching of drilling at lease expiration documented in Herrnstadt, Kellogg and Lewis (2024), to isolate the expiration-induced component of peer activity. The estimator assigns each unique well to one nearby lease and estimates the excess number of wells drilled near expiration. I exclude firm-county-quarters in which the firm's own leases expire in the same county. The estimating equation interacts expiration-driven peer activity with the local network signal to measure how firms' adoption and investment responses vary with signal strength. I estimate these responses separately for each signal and trace their changes across calendar periods and formation-development stages.

At mean signal values, the pooled estimates imply adoption responses ranging from 18 to 35 percent of the mean adoption rate.[7] Peer-technique breadth instead has a negative

[6] See Herrnstadt, Kellogg and Lewis (2024).

[7] Using the mean signal values in Table 1 and a benchmark of 0.50 excess peer wells, the pooled interaction contributions are $0.1531 \times 0.0285 \times 0.50 = 0.00218$ for peer performance and $0.2661 \times 0.0320 \times 0.50 = 0.00426$ for technique intensity. These correspond to 0.22 and 0.43 percentage points, approximately 18 and 35 percent of the mean adoption rate of 1.21 percent.

pooled adoption response. The peer-performance and technique-intensity averages conceal a pronounced timing pattern. Before 2015, the adoption responses to peer performance and technique intensity are weak or negative. Both become positive from 2015 through 2017. After 2017, the responses no longer increase and in some cases decline (Figure 6). This calendar pattern can reflect industry-wide development, accumulated local development or both. Table 3 compares formations at the same stage of local development but different points in the industry-wide transition. The late-minus-early first-block difference is 0.38^{***} for peer performance and 1.29^{**} for technique intensity.[8] The later-block estimates show that the response also varies across stages of local development within each vintage. By 2019, after both vintage groups have reached later blocks, the initial positive difference between late- and early-vintage formations no longer remains.

Adoption and investment provide an additional economic contrast. In early-vintage later blocks through 2014, the technique-intensity response is -0.18^{**} for the probability of a shift toward horizontal drilling but 34.4^{***} for horizontal well counts conditional on positive investment. The contrast separates changes in firms' production technologies from the scale of investment among active investors. Within the model, these opposing responses point to an adoption cost distinct from the return governing additional investment.

I develop a simple model of firms' adoption and investment decisions to interpret the reduced-form pattern. I fit one set of five parameters to ten estimated adoption and investment responses across different stages of industry-wide and local development. The calibrated model reproduces the key empirical comparisons through a combination of return and cost mechanisms whose effects depend on both industry maturity and local development. Removing each component and refitting the remaining parameters causes the model to miss part of that pattern. Economically, the model requires an adoption cost consistent

[8]The late-vintage first-block technique-intensity adoption estimate in Table 3, 0.83, is imprecise on its own. The directly estimated cross-vintage difference is significant because the early-vintage estimate is negative and precise.

with competition for scarce specialized resources. As local experience accumulates, additional local information provides a smaller return benefit and induces a smaller increase in adoption costs. Industry development improves the balance between the return benefit and the adoption cost, making the adoption response more positive. In the calibration, the cost effect declines by more than the return benefit as the industry matures. Initially, the cost effect outweighs the return benefit, producing a negative adoption response even when the investment response is positive. Industry development reduces that cost effect enough for adoption to respond positively but further local development weakens the response as additional information becomes less valuable.

The paper contributes to the literature on investment decisions by showing that peer information can affect a firm's choice of production technology differently from the scale of investment after that choice. Prior studies of peer investment in this setting primarily explain whether and when firms drill. This paper examines their choice of production technology and shows that responses to local information change as the technology and its supporting industry develop. The calibrated model explains how changes in returns and adoption costs can connect this broader development to firms' capital-allocation decisions. This dynamic perspective also connects research on peer information in investment decisions with work on technology diffusion and local learning. Aggregate diffusion models generally abstract from local information. Local-learning research generally treats the broader technology environment as fixed. The paper brings these processes together by showing that firms' responses to local network information at a given stage of local development depend on when that stage occurs in the industry-wide transition.

The paper proceeds as follows. After the literature review, Section 2 describes the setting and the data. Section 3 presents the empirical strategy. Section 4 reports the reduced-form results and what distinguishes the signals. Section 5 reports robustness. Section 6 develops the framework and takes it to the data. Section 7 concludes.

Related Literature

Industry-wide development can change what firms learn from local peers. Research on corporate investment shows that firms use peers' valuations and investment decisions when choosing whether and when to invest (Foucault and Frésard, 2014; Decaire and Wittry, 2024; Bao and Goetz, 2026). Local-learning research similarly finds that peer experience affects technology use when it is informative about the decision-maker's own environment (Foster and Rosenzweig, 1995; Conley and Udry, 2010). These studies establish that peer information matters but generally do not examine whether its value changes as the technology and its supporting industry develop. Growth and diffusion models make that broader development central but generally abstract from information transmitted through local networks (Romer, 1990; Jones, 2022; Aghion and Howitt, 1998; Perla and Tonetti, 2014; Buera and Oberfield, 2020; Stokey, 2021; Comin and Mestieri, 2014). I connect these processes by showing that firms at the same stage of local development respond differently to local information depending on when that development occurs in the industry-wide transition.[9]

This interaction is visible because the paper distinguishes technology adoption from investment scale. Adoption is a fundamental corporate finance choice that determines how a firm produces rather than only how much capital it deploys. Existing finance research on adoption studies how financial conditions, organizational structures and managerial incentives affect the use of new technologies (Ersahin, 2020; Bircan and De Haas, 2020; Mishra, Prabhala and Rajan, 2022; Goehring, Mezzanotti and Ravid, 2026). I instead study how peer-generated information affects adoption and investment together. Their different responses help distinguish changes in the expected return to the technology from changes in the cost of beginning to use it.

Identifying this relationship requires separating information transmitted through a local network from the conditions that produced the network itself. Spatial relationships can

[9] A related corporate-finance literature documents peer effects in other firm policies (Leary and Roberts, 2014).

reflect knowledge spillovers or common geology, input markets and demand conditions. I use spatial models to characterize the strength and content of the existing network rather than to estimate causal peer effects. The measures distinguish information in realized peer performance from information in disclosed implementation choices. Lease-expiration timing then isolates new peer activity within a given network environment. Their interaction estimates whether the same peer activity produces different adoption responses across local information environments.

The oil and gas literature provides the closest empirical comparisons. Kellogg (2011) finds learning within producer-contractor relationships but limited transfer between neighboring producers. Decaire, Gilje and Taillard (2020) shows that real-option exercise responds to peer exercise behavior. Decaire and Wittry (2024) shows that firms delay drilling while waiting to learn from peer projects. Related work studies implementation choices, disclosure, exploration and lease-deadline investment (Covert, 2015; Fetter et al., 2018; Hodgson, 2024; Steck, 2018; Kellogg, 2014; Herrnstadt, Kellogg and Lewis, 2024). These studies primarily explain whether firms drill, when they drill or how productive their wells become. I make the firm's technology mix an outcome, separate technology adoption from drilling investment and study how the response to peer activity changes over the industry-wide transition.

2 Setting and Data

Horizontal drilling with multi-stage hydraulic fracturing is a distinct technology, not simply a more intensive version of conventional drilling. It uses a lateral wellbore and staged fracturing to produce shale and tight-sand resources that vertical wells cannot profitably reach. The well design is fixed when the well is drilled, and each permit records this irreversible technology choice. I use those records to classify the type of well investment and track firm-level adoption. Internet Appendix Figure OA.1 illustrates the two designs.

Oil and gas leases grant firms the right, but not the obligation, to drill specified acreage

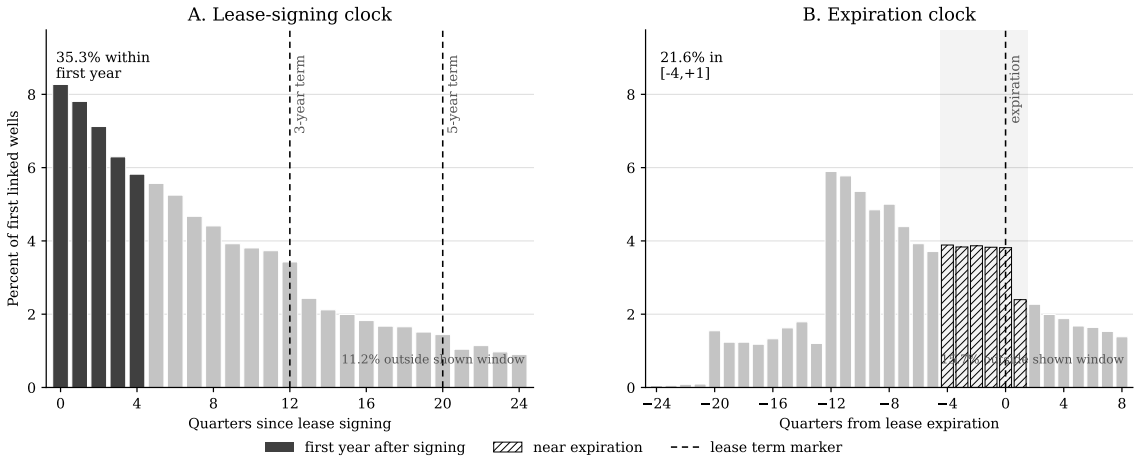


Figure 1: First linked wells around lease signing and expiration

Notes. The unit is the first well linked to a lease under the spatial well–lease match. The main sample uses wells linked to leases within 0.25 miles. Panel A plots event time in quarters since lease signing. Panel B plots event time in quarters relative to lease expiration. Highlighted bars mark the first year after signing and the expiration window $\tau = -4, \ldots, 1$, which is the window used to construct excess peer activity. The vertical axis is the percent of first linked wells. The sample contains 131,610 first linked wells. In Panel A, the dashed vertical lines mark the three-year and five-year primary terms at quarters 12 and 20. Dark gray bars mark the first year after signing, quarters 0 through 4. Hatched white bars in Panel B mark the expiration window.

for a fixed primary term. If no well is drilled before the term expires, the lessee loses those rights. Drilling concentrates immediately after signing and again near contractual expiration (Figure 1). Panel A reports that 35.3 percent of first linked wells occur within the first year after signing, after which drilling generally declines as the lease ages. In Panel B, this decline reverses as expiration approaches. First-well activity rises within four quarters of expiration, and 21.6 percent of first linked wells occur between four quarters before expiration and one quarter after it.

Oil and gas is unusually well suited to measuring how local information affects technology adoption and investment. Mandatory state filings reveal each well's location, timing, technology and design. They identify whether a firm changes its production technology, how many projects it undertakes and the implementation choices disclosed by nearby peers. Production is reported in common physical units, which makes peer performance observ-

able without relying on sales or prices. The resulting project-level data contain adoption, investment scale, disclosed implementation choices and subsequent productivity.

The data come from Enverus, formerly DrillingInfo, a widely used industry data provider that compiles the production, permit, and lease records that states require operators to file.[10] I link each well's monthly output to its permit and lease information through its API number and identify firms by operator. The sample covers onshore wells in the lower-48 states spudded in 2005 or later by firms drilling at least ten wells. A formation is the named geological target recorded for the well, such as the Bakken or Eagle Ford. I use formation in the panel because horizontal productivity and completion choices are comparable within these geological targets but need not be comparable across them. The response panel is at the firm–county–formation–quarter level and spans 2008 through 2022. A county can contain more than one active formation, so county-level peer activity and county controls are measured once at the county-quarter level and attached to each active county–formation row. I also exclude firm–county–quarters in which the operator has same-county leases expiring, so that a firm's response is not confounded with its own drilling deadline. Peer activity is measured at the county level because lease records locate acreage but do not identify the target formation. The county measure also captures nearby drilling outside the 15 formations for which network signals can be estimated. These wells account for about half of expiration-driven peer activity and capture local drilling even when they target smaller, often vertical formations in the same basin. This distinction matters because 105 of the 249 counties contain multiple formations and account for 67 percent of the response panel.

The monthly production records allow me to measure productivity from physical quantities without inferring it from sales, which also reflects prices and demand. Because well output declines steeply with age, productivity is the residual from a standard Arps decline curve. Firm skill is the firm-quarter component of this measure net of location, estimated

[10] These are the same well-completion and production records that underlie EIA's official U.S. production statistics.

using the firm's wells across all formations in which it operates. Appendix B gives both constructions.

The constructed response panel, after excluding firm–county–quarters in which the operator has same-county leases expiring, contains 1,002,519 firm–county–formation–quarter observations, 795 unique firms, 249 unique counties and 15 formations. Table 1 reports descriptive statistics. Adoption and investment are rare at the firm–county–formation–quarter level, so both are zero for most observations.[11] The positive adoption and investment samples used in the intensive specifications contain 12,116 and 14,410 observations, respectively. The all-distance lagged ex-firm horizontal well count averages 22.24, while the lagged expiration-driven peer-activity measure at the baseline 0.25-mile match radius averages 0.049.

I define adoption as an increase in a firm's horizontal share to capture the transition within incumbent operators from vertical to horizontal drilling. The shift toward horizontal drilling has substantial output consequences. The estimated relative-output coefficient for horizontal wells rises from about 1.3 in 2008 to nearly 2.0 by 2019 (Figure 2), so the technology choice affects expected physical output as well as well design. Horizontal wells also rise from about 30 percent of new wells in 2008 to more than 90 percent by 2022 (Figure 3). The same shift occurs among operators active by 2010Q4, not only through entry by new horizontal operators (Figure 4).

Later-developing formations also appear to enter a more standardized engineering environment. Perforated interval length captures well design and project scale, while spud-to-completion time captures execution speed. Among wells in the peer-performance and technique-intensity response samples drilled after 2017, both measures are less dispersed around contemporaneous national practice in later-vintage formations than in early-vintage

[11] Adoption is also bounded from above by the construction in equation (1). In the panel, 120,347 rows have a lagged horizontal share of one and none of them records adoption. The share of rows in that position is not monotone over the episode, at 12.9, 9.9, 11.0 and 14.8 percent in 2008–14, 2015–16, 2017–19 and 2020–21, while the adoption rate is lower in each later period than in the first, at 2.07, 0.93, 1.28 and 0.72 percent. The time path of adoption is not a mechanical consequence of the bound.

Table 1: Descriptive statistics, response panel

Variable	Mean	SD	N
Extensive adoption	0.0121	0.1093	1,002,519
Extensive investment	0.0144	0.1190	1,002,519
Intensive adoption	0.0137	0.9983	1,002,519
Intensive investment	0.0869	1.2364	1,002,519
Expiration-driven peer activity (lag 1)	0.0490	0.2595	1,002,519
Peer performance	0.0285	0.0295	968,821
Peer skill	0.0122	0.0937	968,821
Technique intensity	0.0320	0.0165	788,111
Peer-technique breadth	3.5359	4.3393	1,002,519

Notes. The unit is a firm–county–formation–quarter, restricted to quarters in which the operator has no same-county leases expiring, covering 15 formations over 2008–2022. Extensive adoption is an indicator that the firm's drilling in the county shifts its horizontal share upward, and extensive investment is an indicator that the firm drills any horizontal well. In the response regressions, intensive adoption is the horizontal-share change conditional on positive adoption and intensive investment is the horizontal well count conditional on positive horizontal investment. The intensive-outcome means shown here are unconditional over the displayed sample, with zero values retained when the firm has no positive activity. Intensive adoption is in percentage points of the firm's horizontal share. Peer performance and peer skill are spatial-regression coefficients. Technique intensity is a partial R^2. Peer-technique breadth is a count of attributes. Expiration-driven peer activity is the bunching-weighted measure used in the response regressions, lagged one quarter, constructed at the 0.25-mile lease match radius and excluding the responding firm's own wells. Signal rows have smaller counts where the signal is missing.

formations. Internet Appendix Table OA.1 reports these comparisons. This convergence is consistent with industry-wide learning and standardization in how the technology is implemented. The paper studies how information propagated through local networks affects adoption as this broader industry environment evolves.

The joint adoption and productivity trends define the industry-wide transition used throughout the paper. The transition appears in the joint increase in horizontal adoption and relative productivity and in the displacement of vertical drilling from firms' technology mixes. The reduced-form analysis leaves these changes as separate empirical features instead of collapsing them into a single observed state variable. Calendar periods locate adoption and investment responses within the industry-wide transition, while formation-vintage comparisons separate that timing from a formation's own development. The comparisons reveal whether firms' responses to local peer activity depend on the stage of the broader transition

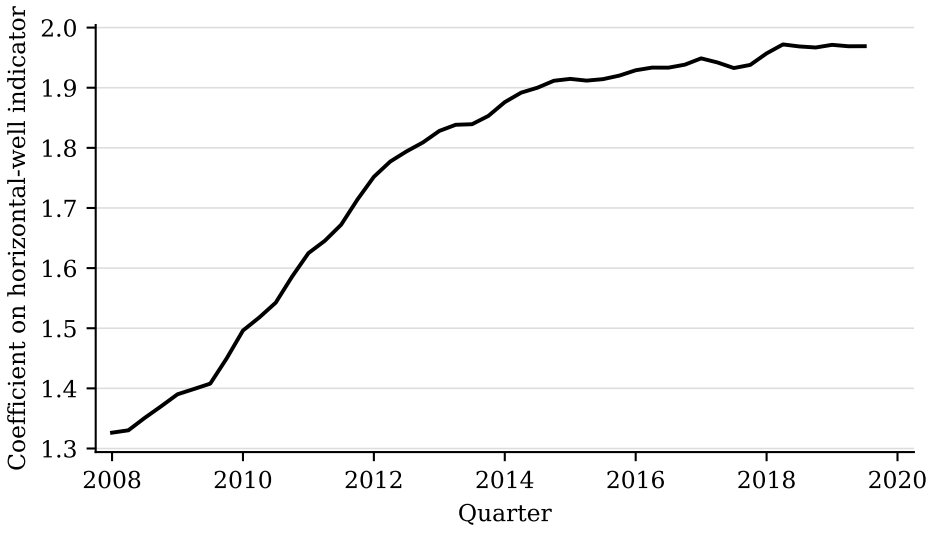


Figure 2: Productivity of horizontal versus vertical wells over time

Notes. The sample contains horizontal and vertical wells drilled in the lower 48 states. The line connects quarterly estimates of γ from 2008 through 2019, each from a separate quarterly well-month regression, $\log(O_{wt}) = \alpha + \beta \log(Age_{wt}) + \gamma Horizontal_w + \varepsilon_{wt}$. O_{wt} is well w's physical output in month t, Age_{wt} is the number of months since production began, and $Horizontal_w$ indicates a horizontal well. The age control accounts for the decline in output as a well matures. Larger values of γ indicate a greater output advantage for horizontal wells.

without assigning that transition to a single mechanism.

3 Empirical Strategy

The empirical strategy estimates how firms' adoption and investment decisions respond to expiration-driven peer activity across local network environments measured from previously completed development blocks. This section defines the peer-activity measure, the local network signals, the outcomes, the response equation and the identifying assumptions.

3.1 Expiration-Driven Peer Activity

Nearby peer drilling is endogenous to the same local returns that shape the firm's own adoption and investment decisions. If peers drill when a formation becomes more attractive,

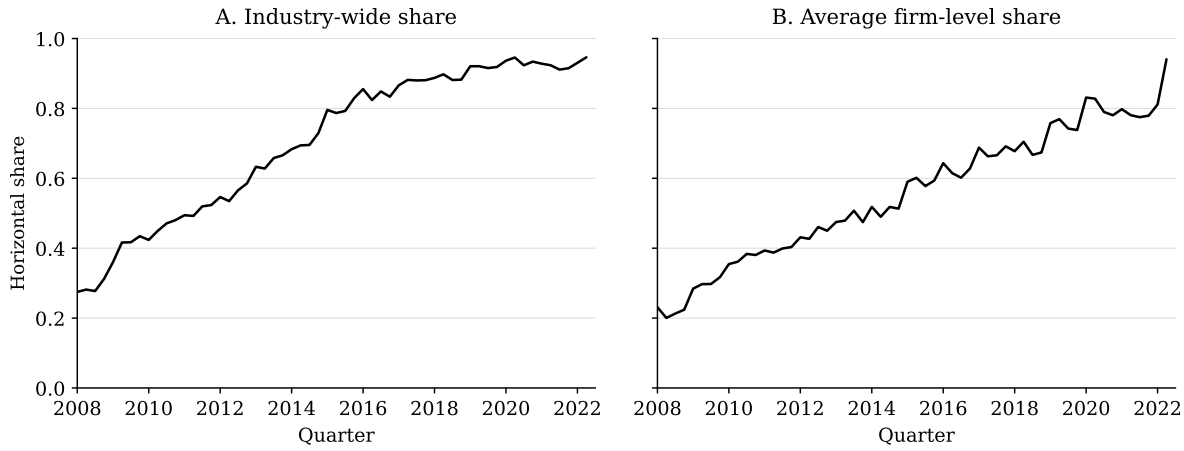


Figure 3: Diffusion of horizontal fracking, 2008Q1 to 2022Q2

Notes. The sample contains wells in the response panel from 2008Q1 through 2022Q2. The left panel divides the total number of horizontal wells drilled in each quarter by the total number of horizontal and vertical wells. The right panel first computes this horizontal share separately for every operator that drills in the quarter, then reports the unweighted average across those operators. The left panel weights operators by drilling activity, while the right panel gives each active operator equal weight.

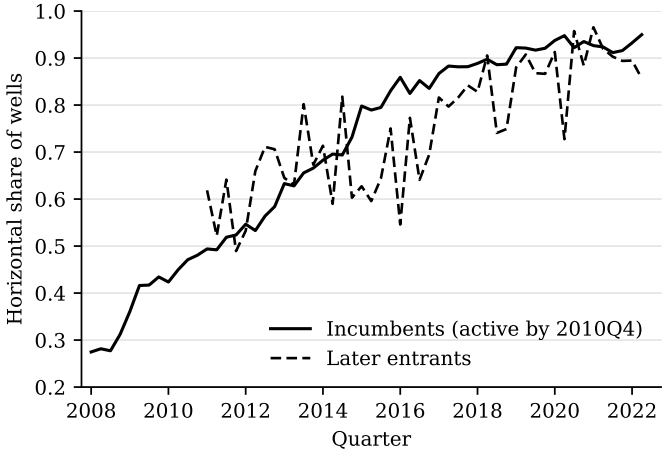


Figure 4: Horizontal diffusion among incumbents versus later entrants

Notes. The sample contains wells in the response panel from 2008Q1 through 2022Q2. Incumbents are operators whose first active quarter was 2010Q4 or earlier. Later entrants are operators whose first active quarter occurred after 2010Q4. For each group and quarter, the horizontal share equals the group's total horizontal wells divided by its total horizontal and vertical wells. The series are group-level shares, not averages of firm-level shares.

a raw peer-activity measure combines local evidence with the common shock moving all firms in the area. The design uses lease expirations to isolate peer drilling whose timing is shifted by a contractual deadline set years earlier. Oil and gas leases typically expire unless the lessee drills before the end of the primary term. A well drilled near expiration is more likely to reflect the lease deadline than contemporaneous local returns.

The construction begins by assigning each unique well to one candidate lease. Eligible leases must be signed before the well is spudded and lie within 0.25 miles of the well. I assign the nearest lease and use the earliest expiration to break ties among leases at nearly the same distance. The spatial assignment is inferred, not an observed legal link between a well and a lease. It permits the well operator and leaseholder to have different standardized names. Section 5.4 rebuilds the measure using operator–lessee name agreement and reports the corresponding response estimates.

I then measure event time as the number of quarters from the assigned lease's expiration to the well's spud date. Assigned wells bunch near lease expiration under the baseline spatial match (Figure 5). I fit a smooth fifth-order event-time profile over quarters -12 through 4, with separate indicators for quarters -4 through 1. Setting those indicators to zero gives the counterfactual count in Figure 5. In both the quarter before expiration and the expiration quarter, about 32 percent of observed assigned wells lie above the fitted counterfactual. Each assigned well receives the estimated excess share at its event time as a fractional weight, with zero weight outside the expiration window. Appendix A gives the estimator and the excess-share calculation.

The peer-activity measure sums these weights within a county-quarter. I remove the responding firm's own operated and leased wells, then lag the resulting measure one quarter.[12] I also exclude firm-county-quarter observations in which the firm has its own same-county lease expiring in the response quarter. Section 5.1 widens this exclusion to the full bunching window. The restriction prevents the measure from substituting the firm's own lease deadline

[12]Section 5.4 reports additional lags.

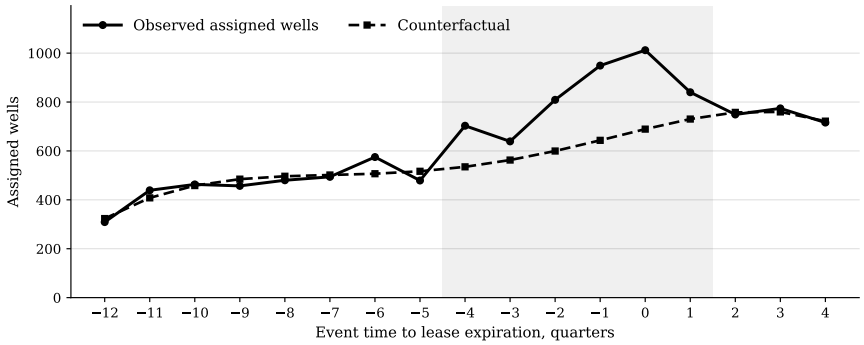


Figure 5: Lease-expiration bunching in assigned wells

Notes. The unit is a unique well assigned to the nearest eligible lease signed before the well was drilled and located within 0.25 miles. Event time is the number of quarters between the well's drilling date and the assigned lease's expiration. The figure plots the observed number of wells at each event time and the fitted counterfactual from $C_\tau = \alpha + \sum_{p=1}^{5} \eta_p \tau^p + \sum_{k=-4}^{1} \lambda_k \mathbf{1}\{\tau = k\} + u_\tau$, estimated over quarters −12 through 4. The counterfactual sets the expiration-window indicators to zero. The share of observed wells above that counterfactual at each event time, bounded between zero and one and set to zero outside quarters −4 through 1, is the fractional weight assigned to each well when constructing expiration-driven peer activity. Observed counts are the black solid line with circles. The fitted counterfactual is the black dashed line with squares. The shaded band marks the expiration window, event quarters −4 through 1.

for prior peer activity.[13] Internet Appendix Figure OA.2 reports the distribution of the resulting measure across response samples and formations.

3.2 Measuring Local Network Signals

Local network signals classify the network environment represented by earlier peer wells. I estimate them separately within the geological formations that define the response panel in Section 2.[14] To make the signals dynamic, I sort horizontal wells in each formation by first production month, split them into consecutive 2,000-well blocks, and assign each firm-county-formation-quarter the signal from the most recently completed block in that formation.

The block precedes the response quarter, but its productivity measure uses the full sub-

[13]Section 5.1 reports the sample loss from both versions of this exclusion.

[14]Formation labels are simplifications. Some named formations contain multiple subformations. Finer geological units would be more precise in principle but would often leave too few wells to estimate the network measures reliably.

sequent production record of the wells in that block. The signals are ex post classifications of the network environment, not literal measures of the information available to firms when the block closed.

The signal construction uses formation-block spatial regressions.[15] The baseline model relates well productivity to weighted productivity and operator skill from other firms' wells that began producing earlier. Additional specifications replace operator skill with exposure to peers' disclosed drilling and completion attributes. The spatial specifications produce the four local network signals described below. The weight matrix W determines which earlier peer wells enter each measure. I estimate the signals under alternative weights that emphasize geographic proximity, recency, a common development episode, or relative operator skill. Appendix C gives the full specifications and weight definitions.

Peer-performance networks capture local performance clustering. In each formation-block, the spatial regression relates a well's productivity to weighted productivity from other firms' wells that began producing earlier. The resulting coefficient measures the strength of local productivity comovement. The source of that clustering remains open. Shared geology, service markets, operating constraints, copying, skill transfer, or a portable operating recipe can all contribute. The same regression includes the weighted skill of nearby peer operators. The peer-skill coefficient measures whether nearby operator skill has an additional association with productivity after accounting for peer performance. Because skilled operators are also likely to drill productive wells, I use peer skill as a diagnostic, not a separate network channel.

The two peer-technique signals use disclosed drilling and completion attributes, including fracking inputs and pressure readings. For each attribute, I estimate whether a well's productivity is associated with exposure to earlier peers whose records report that attribute, conditional on peer performance and the spatial-model controls. The resulting coefficient

[15]Spatial relationships may reflect both peer interactions and common environments (Anselin, 2001; Manski, 1993). Network structure can sometimes help separate these forces (Bramoullé, Djebbari and Fortin, 2009). Applications to technology adoption include Conley and Udry (2010) and Bandiera and Rasul (2006).

compares wells with greater and lesser exposure to peers for whom that information is available. Because disclosure rules, reporting systems, and compliance vary across states and over time, the measure cannot separate information availability from the usefulness of the information disclosed. Peer-technique breadth counts the attributes with positive and statistically precise productivity loadings. Technique intensity is the additional explanatory power from including all peer-technique exposure terms in the spatial model. The four local network signals enter the response regressions in their original units. Coefficients can be compared over time for the same signal and outcome, but not across signals.[16] Appendix D lists the attributes and gives the peer-technique regressions.

3.3 Outcomes and Samples

I distinguish adoption from investment using four outcomes. Adoption records whether drilling in a county-formation-quarter shifts the firm's overall technology mix toward horizontal wells. Investment records whether and how many horizontal wells the firm drills in that market. Let H_{iq} and V_{iq} denote firm i's accumulated horizontal and vertical wells at the start of quarter q, across all markets in which it operates. Let h_{icgq} and v_{icgq} denote its horizontal and vertical wells in county c, formation g, and quarter q. The percentage-point change in the firm's horizontal share is

$$A_{icgq} = 100 \left(\frac{H_{iq} + h_{icgq}}{H_{iq} + V_{iq} + h_{icgq} + v_{icgq}} - \frac{H_{iq}}{H_{iq} + V_{iq}} \right). \tag{1}$$

I define extensive adoption as an indicator that $A_{icgq} > 0$. Conditional on a positive shift, A_{icgq} measures intensive adoption. Firms can shift their technology mix toward horizontal drilling over multiple quarters. The adoption measure records each increase in horizontal share, not only a firm's first horizontal well. Section 5.3 examines whether the estimated

[16] Related work uses disclosed drilling and completion information to study social learning and productivity in hydraulic fracturing (Covert, 2015; Fetter et al., 2018; Steck, 2018).

effects vary with prior horizontal experience. Extensive investment indicates that $h_{icgq} > 0$. Conditional on positive horizontal investment, h_{icgq} measures the intensive investment level. Because each intensive sample conditions on the corresponding extensive outcome, comparisons between extensive and intensive coefficients are suggestive and do not compare a common sample directly.

Equation (1) separates adoption from investment, which is why horizontal drilling alone is insufficient as an outcome. A firm that already drills only horizontal wells can drill a great many more without changing its technology mix at all and it registers zero adoption however large its investment. A firm that drills one horizontal well while half its stock is vertical registers a positive shift. The measure asks whether this market's drilling moved the firm toward the new technology, not how much drilling occurred, and it is a firm-level object that county drilling counts cannot deliver.

3.4 Main Empirical Framework

I estimate the response in the firm-county-formation-quarter panel from 2008 through 2022:

$$Y_{icgq} = \beta \left(\widehat{E}^{-i}_{c,q-1} \cdot S_{gq} \right) + \theta_E \, \widehat{E}^{-i}_{c,q-1} + \theta_S \, S_{gq} + \mu_i + \kappa_c + \alpha_{ic} + \lambda_q + X'_{c,q-1}\Gamma + X^{f\,\prime}_{iq}\Phi + u_{icgq}. \quad (2)$$

Here Y_{icgq} is one of the four outcomes defined above. The variable $\widehat{E}^{-i}_{c,q-1}$ is the estimated number of expiration-driven peer wells in county c during the previous quarter after removing wells operated or leased by firm i. Each assigned well contributes the excess-share weight for its event time shown in Figure 5, and these weights are summed within the county-quarter. The variable S_{gq} is the local network signal from the most recently completed block in formation g. The coefficient β measures how the response to one additional estimated excess peer well changes when the local signal increases by one unit. The coefficients θ_E and θ_S include the corresponding lower-order terms.

The regressions absorb operator fixed effects μ_i, county fixed effects κ_c, operator-by-

county fixed effects α_{ic}, and calendar-quarter fixed effects λ_q. Firm-county fixed effects compare a firm's outcomes within the same county over time and remove time-invariant differences across its local operations. Calendar-quarter fixed effects remove shocks common to all counties in a quarter, including oil prices and aggregate drilling conditions. Lagged county controls account for changes in local drilling activity and other county-level conditions. They are the county horizontal-production share and annual county size. The firm-quarter controls X_{iq}^{f} are the firm's horizontal-production share, its horizontal and total well stocks, its county-level horizontal well stock, and a firm-quarter productivity index. Annual county size is the county's average quarterly well count over the calendar year. The firm-quarter productivity index is the firm-quarter component of well productivity described in Appendix B, standardized within each quarter. Standard errors are clustered by formation-block. Unless otherwise noted, Section 4 uses this specification.

3.5 Specification across Eras

I allow the coefficients on the controls to vary across eras. The firms and formations active early differ from those active later, so imposing a common relationship between the controls and outcomes across the full period can distort changes in the estimated peer-activity response. The preferred specification estimates equation (2) separately within each era. An alternative specification holds the control coefficients constant across eras. The two approaches differ most clearly for the early technique-intensity estimate, while their later estimates are similar. Internet Appendix Table OA.12 reports both specifications. All differences across eras are tested directly.

3.6 Identification and Threats

Identification requires expiration-driven peer activity to be unrelated to unobserved changes in the responding firm's outcome after conditioning on the lower-order terms, fixed effects

and controls in equation (2). The conditional-mean assumption applies to the variation in peer activity used to estimate adoption and investment effects within each measured network state. The network signals need not be exogenous to local development because the specification targets heterogeneity in the effect of peer activity, not the causal effect of changing a signal. They identify where the adoption or investment response to expiration-driven peer activity is stronger or weaker. Their construction and timing still create separate threats discussed below.

The first threat is the responding firm's own expiration pressure. The peer-activity measure removes wells operated or leased by the responding firm. The baseline sample also excludes firm-county-quarters in which the firm has a lease expiring in the same county. An expiration elsewhere in the firm's portfolio could still cause it to mobilize drilling across several markets. Section 5.1 applies a stricter exclusion to every firm-quarter in which the firm has a matched lease expiration anywhere in its portfolio. The main sign pattern remains under this restriction.

The second threat is that the bunching estimator may classify drilling driven by local conditions as expiration-driven. Drilling declines after lease signing and then increases again as expiration approaches (Figure 1). The reversal is difficult to reconcile with a smooth local development trajectory alone. The estimator defines excess drilling relative to the event-time profile outside the expiration window and assigns each well only the estimated excess share at its event time. The expiration spike remains when the bunching specification uses a wider sample of matched wells and when wells are linked to leases using operator-lessee name matches. The name-matched response estimates also retain the main pooled sign pattern. Internet Appendix Tables OA.2–OA.3 report these analyses. Internet Appendix Table OA.4 addresses uncertainty from the estimated peer-activity measure. The county bootstrap re-estimates the bunching curve, reconstructs peer activity and re-estimates the response equation in each draw while holding the network signals fixed. The bootstrap percentile intervals preserve the central pooled results for extensive adoption and exclude zero

for technique intensity on intensive investment. The bootstrap standard errors exceed the formation-block-clustered standard errors in seven of the eight reported specifications. The exception is technique intensity on intensive investment. The breadth estimate for intensive investment loses precision. Expiration-driven drilling also reflects a leaseholder's choice to drill instead of letting the lease lapse. Section 5.1 finds no sign that this choice selects on favorable local conditions. New leasing fails to rise with expiration-driven drilling. Wells drilled in the expiration window have no productivity advantage where more expiring leases are drilled. Among county-quarters with scheduled expirations, positive recent productivity innovations are associated with slightly less expiration-driven drilling.

The network signals raise separate measurement and interpretation concerns. They are estimated relationships among earlier wells, not measures of productivity levels. Peer performance is the coefficient on weighted prior peer productivity, while breadth and intensity summarize how disclosed peer techniques relate to productivity. Because the outcomes are adoption and investment, the dependent variable does not enter the signal construction mechanically. The signals may still proxy for geology, local productivity or development maturity. Formations with declining peer-performance coefficients continue to experience strong productivity growth after accounting for the lower-order signal term in the response equation. Internet Appendix Section OA.4.2.1 and Figure OA.3 report the corresponding trajectory analysis. A declining network coefficient is not equivalent to declining productivity. Web Appendix Table W.1 reports the four preferred network measures for every completed formation block. Internet Appendix Table OA.5 compares alternative spatial weights on the same blocks. The comparison separates sensitivity to spatial weights from differences in the stage of local development. The analyses clarify the variation used by the interaction without making the signals causal.

Signal timing creates an additional concern. Each signal is estimated from a completed 2,000-well formation block and remains fixed until the next block closes. A timing analysis drops the first response quarter after every block closure, removing the quarter in which

wells used to complete the block could also enter lagged peer activity. The headline pooled signs remain in Internet Appendix Table OA.6. Equal-sized blocks hold the number of wells used to estimate each network state approximately constant across updates. Appendix C gives the block construction. The reported standard errors nevertheless treat the estimated network signals as fixed.

A remaining threat is that formation discovery produces a wave of lease signing, followed by clustered expirations several years later. The locations of those expirations are not random. Their timing is less likely to reflect current information about the formation. Among first linked wells, 35.3 percent are drilled within the first year after signing. First-well activity then declines before increasing again among leases that remain as expiration approaches (Figure 1). An earlier discovery may explain why many leases were signed in the same place, but it is less likely to explain why drilling on the remaining leases increases in the expiration quarter instead of during the intervening years. The bunching estimator assigns only this incremental expiration-related share to peer activity. Removing the responding firm's wells and expiration exposure then isolates its response to drilling shifted by its peers' deadlines. Calendar-quarter effects absorb the common oil-price level and other industry-wide conditions. Oil prices could still change firms' sensitivity to peer activity or network signals, so Section 5.2 interacts prices with both. The main adoption estimates remain similar and the price interactions are imprecise. To distinguish a response specific to horizontal technology from a general increase in local drilling, I re-estimate equation (2) with vertical drilling as the outcome. A general local drilling boom should produce a parallel positive vertical response. No such pattern appears in the estimates. Section 5.4 varies the lag and lease-matching distance to determine whether peer-activity effects are concentrated at the temporal and spatial ranges used in the baseline design. An unobserved local shock could still coincide with clustered expirations and affect firms differently across network states. I interpret the estimates as net responses to expiration-driven peer activity, not as pure information effects.

4 Local Network Responses to Expiration-Driven Peer Activity

Firms' adoption and investment responses to expiration-driven peer activity depend on the local network signal and the timing of local development within the industry-wide transition. I first report pooled responses, then trace adoption over calendar periods and finally compare formation vintages at common stages of local development. The four outcomes distinguish shifts in firms' technology mixes from whether and how much they invest horizontally.

4.1 Pooled Network Responses

Table 2 reports the baseline response pooled across the whole sample period. The four columns separate adoption from investment, each on the extensive and intensive margin. Peer performance is positive and significant for extensive adoption, extensive investment and intensive adoption. Technique intensity is positive and significant for extensive adoption and extensive investment. Its intensive estimates are positive but imprecise. Peer-technique breadth has the opposite effect, with negative interactions where it is precise. Peer skill is imprecise for both extensive outcomes and intensive adoption. Its negative intensive-investment estimate is not robust to the county bootstrap in Internet Appendix Table OA.4. Because the signal measures use different units, I do not compare raw coefficient magnitudes across signals. Web Appendix Table W.3 reports descriptive statistics for the peer-performance and technique-intensity response samples used in the magnitude comparisons below.

To interpret the magnitudes, consider Haynesville as a concrete example. Haynesville's peer-performance signal is close to the average and its technique-intensity signal is above the average. Its peer-performance signal is 0.044, and positive expiration-driven peer activity averages 0.50 excess wells. Applying the pooled extensive-adoption coefficient of 0.153^{**} implies a 0.33 percentage point larger probability that the firm's drilling shifts its horizontal

Table 2: Pooled adoption and investment responses to expiration-driven peer activity, by local network signal

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance [distance-time]				
$\widehat{E}^{-i}_{c,q-1}\times$ signal	0.1531**	0.1835**	9.4487***	5.6188
	(0.0649)	(0.0756)	(2.6951)	(6.6970)
N	582,531	582,531	7,234	8,565
R^2	0.381	0.402	0.911	0.641
Peer skill [distance-time]				
$\widehat{E}^{-i}_{c,q-1}\times$ signal	0.0404	0.0272	−1.7864	−37.4633**
	(0.1688)	(0.1014)	(2.3706)	(15.5702)
N	152,297	152,297	912	1,080
R^2	0.536	0.541	0.980	0.866
Peer-technique breadth [equidistant]				
$\widehat{E}^{-i}_{c,q-1}\times$ signal	−0.0020***	−0.0025***	−0.0222	−0.3267*
	(0.0006)	(0.0007)	(0.0367)	(0.1741)
N	683,440	683,440	7,058	8,420
R^2	0.354	0.370	0.957	0.579
Technique intensity [distance-time]				
$\widehat{E}^{-i}_{c,q-1}\times$ signal	0.2661***	0.2988**	8.8524	37.6117
	(0.0955)	(0.1249)	(7.3287)	(27.1502)
N	778,002	778,002	8,704	10,348
R^2	0.358	0.378	0.862	0.615
Firm-quarter controls	Yes	Yes	Yes	Yes
County-quarter controls	Yes	Yes	Yes	Yes
Operator / county / operator-county / quarter FE	Yes	Yes	Yes	Yes
Standard errors	formation-block	formation-block	formation-block	formation-block

Notes. The unit is a firm–county–formation–quarter. The sample excludes firm–county–quarters in which the responding firm has a lease expiring in the same county. Each entry reports the coefficient on one-quarter-lagged expiration-driven peer activity, $\widehat{E}^{-i}_{c,q-1}$, interacted with the listed network signal, S_{gq}, from a separate regression for the indicated outcome. Extensive adoption indicates a positive change in the firm's horizontal share, $A_{icgq} > 0$. Intensive adoption is A_{icgq} conditional on a positive change, in percentage points. Extensive investment indicates at least one horizontal well. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Each signal-specific sample contains observations for which that signal is positive and statistically different from zero under its preferred weight. The intensive-outcome samples also require a positive value of the corresponding outcome. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Brackets give the spatial weight used to estimate the signal. Coefficients are the change in the outcome per estimated excess peer well per unit of the signal. Extensive outcomes are probabilities. Standard errors are clustered by formation-block and reported in parentheses. N and R^2 are reported for each regression. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

share upward. The implied change is about 27 percent of the average extensive-adoption rate of 1.21 percent in Table 1. The intensive-adoption coefficient of 9.45^{***} implies a 0.20 percentage point larger horizontal-share shift, about 17 percent of a typical positive adoption change.[17] The extensive-investment response is also positive, while the intensive-investment estimate is positive but imprecise.

At the Haynesville benchmark, the technique-intensity signal is 0.047. The pooled coefficients imply a 0.63 percentage point increase in the probability of adoption. The intensive-investment point estimate implies 0.88 additional horizontal wells conditional on positive horizontal investment, but it is imprecise under formation-block clustering. Peer performance likewise has a clear adoption effect but no precise intensive-investment response.[18] Across both signals, the pooled estimates provide stronger evidence that peer activity changes adoption and the decision to invest than that it changes investment levels conditional on positive investment. They are consistent with local productivity comovement helping firms assess whether the technology is viable under shared local conditions and with observed peer techniques informing implementation, but they do not distinguish learning from shared geology, services, or other local complements.

Technique intensity is the partial R^2 from jointly adding the peer-technique terms, so greater attribute availability can mechanically increase its explanatory power. The measure remains informative because firms can use only techniques they observe. Reporting volume produces no detectable heterogeneity in the peer-activity effect across the full panel (Internet Appendix Table OA.8). Within formation blocks where technique intensity is positive and precisely estimated, reporting volume and intensity move together, so the two cannot be separated empirically. Technique intensity is best read as combining the availability of peer data with the extent to which those data jointly explain productivity. It does not isolate

[17] Internet Appendix Table OA.7 reports standardized estimates where peer activity and network signals are scaled by one standard deviation.

[18] The estimates use signal-specific samples, and I do not test the difference between the peer-performance and technique-intensity estimates.

informativeness holding availability fixed.[19]

Peer-technique breadth presents a sharp contrast. Internet Appendix Table OA.9 groups formations according to whether breadth increases or remains flat across successive 2,000-well development blocks. Formations with increasing breadth add more attributes with positive productivity loadings but those attributes overlap more in what they explain. Greater breadth may characterize formations where productive drilling requires a larger or more complex set of techniques instead of providing a clearer guide to adoption. If breadth reflects a more demanding implementation problem, additional attributes may attenuate the adoption response to peer activity instead of strengthening it. I examine the design of firms' own wells to assess whether breadth matters after adoption. Perforated interval measures the length of the completed horizontal section and provides a proxy for project scale and complexity, while drill time provides a measure of execution speed and, less directly, cost. Breadth is associated with longer perforated intervals at lags two and three in Internet Appendix Table OA.10 but has no consistent relationship with drill time. The additional technique information may help firms already using the technology configure more complex wells while revealing a higher implementation threshold for firms deciding whether to adopt. The combined evidence suggests that peer activity affects different decisions depending on what the local network reveals. Performance comovement bears on local viability, jointly informative techniques bear on implementation and scale, and a broad but overlapping set of techniques bears on project design while potentially increasing the threshold for adoption.

4.2 The Adoption Response over Time

The pooled estimates combine periods in which horizontal adoption, productivity, and supporting industry capacity differ substantially. The adoption response changes across calendar

[19]This interpretation differs from work that treats disclosure primarily as expanding the information available to peers. Here the distinction is between the amount of disclosed material and its joint relationship with productivity.

periods (Figure 6). Before 2015, the adoption effect is not positive even where the measured local signal is positive and statistically precise. The interaction is -0.046 and imprecise for peer performance and -0.174^{***} for technique intensity. It becomes positive for both signals in 2015 to 2017, at 0.282^{***} and 0.357^{*}, respectively. After 2017, the peer-performance response falls while the technique-intensity response remains positive.[20]

Direct tests confirm the reversal. Between the first two periods, the peer-performance response rises by 0.328^{***} and the technique-intensity response rises by 0.531^{**}. After 2017 the two estimates follow different paths. The peer-performance response falls by 0.610^{***}, while the technique-intensity change is small and imprecise.[21]

Adoption and investment can also move in opposite directions (Figure 6). From 2008 through 2014, the technique-intensity interaction is -0.174^{***} for adoption and 29.3 for investment levels conditional on positive horizontal investment, although the investment estimate is imprecise. The early negative adoption response is absent from the intensive investment level. Section 6 uses this difference and the later divergence between peer performance and technique intensity to distinguish among possible mechanisms. Before turning to those mechanisms, the next subsection separates calendar timing from accumulated local development.

4.3 Formation Vintage and Local Development

The reversal in Figure 6 could reflect accumulated local development, when that development occurs within the industry-wide transition, or both. Table 3 separates these dimensions using the quarter in which a formation completes its first 2,000-well block. Early-vintage

[20]Internet Appendix Table OA.11 enters lags one through four jointly for peer performance and technique intensity on extensive adoption and intensive investment. The added terms indicate some persistence but leave the one-quarter lag interpretation unchanged.

[21]The signals use different estimation samples, so I do not test whether the rise and fall contrasts differ across the two signals. Internet Appendix Table OA.13 tabulates intensive investment for all three signals, the two margins not plotted and the peer-technique-breadth estimates. Web Appendix Table W.5 reports the complete four-margin table.

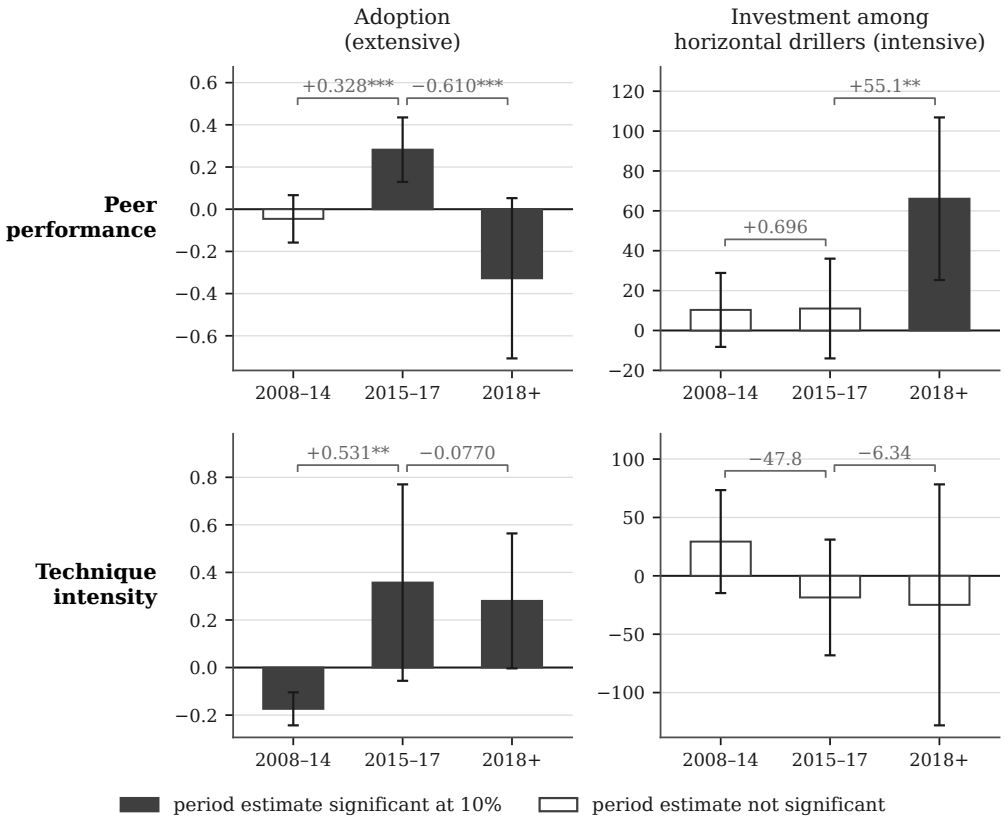


Figure 6: Adoption and intensive-investment responses by period

Notes. The unit is a firm–county–formation–quarter. Each bar reports the coefficient on one-quarter-lagged expiration-driven peer activity interacted with peer performance (top row) or technique intensity (bottom row). The left column uses extensive adoption. The right column uses horizontal well counts among firms with positive horizontal investment. Each regression is estimated separately for 2008–14, 2015–17 or 2018 onward. It includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Control coefficients are allowed to differ across periods. Both signals use distance-time weights. Each signal-specific sample contains observations for which that signal is positive and statistically different from zero. The brackets report directly estimated contrasts between adjacent periods. Whiskers are 95 percent confidence intervals based on formation-block-clustered standard errors. Filled bars are significant at the 10 percent level. The vertical axis reports coefficients. In the left column, a coefficient is the change in the probability of extensive adoption per estimated excess peer well per unit of the signal. In the right column, it is the change in the number of horizontal wells per estimated excess peer well per unit of the signal. Internet Appendix Table OA.13 tabulates intensive investment for all three signals and reports the additional margins and peer-technique-breadth estimates. Web Appendix Table W.5 reports all three signals and all four outcomes. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

formations complete that block by 2012Q3. Late-vintage formations complete it in 2014Q1 or later. No formation completes its first block between these dates. The first-block estimates are not restricted to a single calendar period. They compare formations according to when the first local-development state was established. The later-block estimates are divided by response period. The saturated specification estimates each vintage-period response jointly and reports direct cross-vintage differences where both groups are observed. Internet Appendix Table OA.27 reports the aligned intensive-investment counterpart, including the common 2019–21 comparison.

Both signals differ across vintages in the first-block state. For peer performance, the adoption response is -0.14^{*} in early-vintage formations and 0.23^{**} in late-vintage formations. The late-minus-early difference is 0.38^{***}. For technique intensity, the corresponding estimates are -0.46^{**} and 0.83. Although the late-vintage coefficient is imprecise, the directly estimated difference is 1.29^{**}. The evidence lies in the change across formation vintages, not a separately precise positive technique-intensity response in late-vintage first blocks.

The subsequent technique-intensity estimates trace how that initial difference changes as local development accumulates. Among early-vintage formations in later blocks, the adoption response moves from -0.18^{**} through 2014 to 0.19^{***} in 2015–16. The difference is 0.37^{***}. The estimate is 0.16 from 2017 onward and 0.18 from 2019 onward. Late-vintage formations do not reach later blocks until 2017. Their technique-intensity response is 0.27^{*} from 2017 onward and 0.03 from 2019 onward. Over the common 2019–21 period, the late-minus-early difference is -0.15 and imprecise. The initial cross-vintage difference in technique intensity dissipates once both groups have accumulated local development.

Peer performance follows a different later path. From 2019 onward, its adoption response is 0.28 in early-vintage later blocks and -0.78^{**} in late-vintage later blocks. The late-minus-early difference is -1.07^{**}. The negative difference cannot be attributed to greater accumulated development in the late-vintage group. Early-vintage observations are further along on average, at approximately block 4.6 compared with block 3.5 for late-vintage observations.

Table 3: Extensive-adoption response to expiration-driven peer activity by formation vintage, block position and calendar timing

Era and local-development state	Early vintage	Late vintage	Late minus early
Panel A. Peer performance, extensive adoption			
First-block state	-0.144^{*}	0.234^{**}	0.378^{***}
	(0.077)	(0.093)	(0.120)
Later blocks, through 2014	-0.069	—	—
	(0.065)		
Later blocks, 2015–16	-0.133	—	—
	(0.232)		
Later blocks, 2017+	0.227	-0.677^{*}	-0.904^{**}
	(0.207)	(0.348)	(0.405)
Later blocks, 2019+	0.284	-0.782^{**}	-1.066^{**}
	(0.316)	(0.306)	(0.440)
Panel B. Technique intensity, extensive adoption			
First-block state	-0.460^{**}	0.826	1.286^{**}
	(0.228)	(0.591)	(0.634)
Later blocks, through 2014	-0.184^{**}	—	—
	(0.073)		
Later blocks, 2015–16	0.188^{***}	—	—
	(0.060)		
2015–16 minus through 2014	0.372^{***}	—	—
	(0.081)		
Later blocks, 2017+	0.157	0.267^{*}	0.110
	(0.195)	(0.145)	(0.243)
Later blocks, 2019+	0.183	0.032	-0.150
	(0.241)	(0.360)	(0.434)
Firm-quarter controls	Yes	Yes	Yes
County-quarter controls	Yes	Yes	Yes
Operator / county / operator-county / quarter FE	Yes	Yes	Yes
Standard errors		formation-block	

Notes. The unit is a firm–county–formation–quarter. The sample excludes firm–county–quarters in which the responding firm has a lease expiring in the same county. Each entry is the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed signal in an extensive-adoption regression. Coefficients are the change in the probability of adoption per estimated excess peer well per unit of the signal. Early-vintage formations complete their first 2,000-well development block by 2012Q3. Late-vintage formations complete it in 2014Q1 or later. First-block rows use observations whose signal comes from the formation's first completed block, with no restriction on the response period. Period labels refer to the calendar quarter of the response observation. Later blocks contain blocks two and above. The 2019+ row is nested within the 2017+ row. Dashes indicate that late-vintage formations have no later-block observations before 2017. The first-block through 2017+ rows come from a saturated specification that jointly estimates the listed vintage-period responses. The 2019+ row re-estimates the aligned later-block comparison over 2019–21. The last column reports directly estimated late-minus-early differences. The additional technique-intensity row reports the directly estimated change within early-vintage formations from later blocks through 2014 to later blocks in 2015–16. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Both signals use distance-time weights. Each signal-specific sample contains observations for which that signal is positive and statistically different from zero. Standard errors are clustered by formation-block and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Local information saturation alone would predict the opposite ordering.

The adoption and investment estimates provide additional evidence on how to interpret the vintage patterns. In early-vintage later blocks through 2014, the technique-intensity response is -0.18^{**} for adoption but 34.4^{***} for horizontal-well investment conditional on positive investment. Peer performance is imprecise for adoption in the same group but has an intensive-investment response of 35.4^{**}. In late-vintage later blocks from 2017 onward, the peer-performance response is -0.68^{*} for adoption and 114.3^{*} for intensive investment. From 2019 onward, the technique-intensity response is close to zero for adoption but 203.2^{***} for intensive investment. Local signals can affect investment levels conditional on positive investment after their adoption response has faded or become negative.

The first-block and later-block comparisons reveal how industry timing and local development interact. A purely local account would predict the same first-block response across formation vintages, contrary to the first-block differences for both signals. One interpretation is that local development has two offsetting effects early in the transition. Additional local information becomes less valuable but accumulated development also reduces the cost of acting on that information. When local development begins later, those starting costs may already be lower, leaving information saturation as the dominant force. Section 6 formalizes this interaction.

The two signals point to different economic mechanisms. The technique-intensity pattern is consistent with industry-wide development making local technique information easier or less costly to use when a formation first develops, followed by diminishing informational value as local experience accumulates. Peer performance need not follow the same path because realized outcomes can convey both local opportunities and the competitive consequences of successful nearby activity. Its negative adoption response in late-vintage later blocks, combined with positive intensive-investment responses, could reflect competition deterring remaining nonadopters, higher investment levels among firms with positive horizontal in-

vestment, or selection in the firms that remain at risk of adoption.[22] I leave distinguishing among these explanations to future research.

5 Robustness

The robustness analysis addresses four threats to the interpretation. Subsection 5.1 examines the identifying assumption that expiration timing shifts peer drilling independently of the responding firm's own incentives to adopt or invest. Subsection 5.2 considers whether oil prices or fixed differences across formations explain the calendar and block-vintage patterns. Subsection 5.3 evaluates alternative explanations for adoption and investment, including general drilling booms, responses without measured network signals, competition and changes in firm composition. Subsection 5.4 examines whether the results depend on how peer activity and network signals are constructed and timed.

5.1 Endogeneity of Expiration-Driven Drilling

The design uses expiration timing as a source of variation in peer drilling. The identifying assumption is that this timing is unrelated to unobserved changes in the responding firm's incentives to adopt or invest. The responding firm's own leases may expire alongside its peers' leases, so the baseline sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. An expiration elsewhere could induce drilling across the firm's portfolio, including in counties where it has no lease expiring. Internet Appendix Table OA.14 accordingly excludes every firm-quarter in which the firm has a matched expiration anywhere in its portfolio and also excludes observations with an own same-county expiration anywhere in the full six-quarter bunching window. The pooled and block-vintage

[22] Information spillovers and competitive effects can coexist (Bloom, Schankerman and Van Reenen, 2013). Recent evidence also finds that local peer investment can reduce firms' investment when nearby investments are strategic substitutes (Bao and Goetz, 2026).

patterns remain similar under both restrictions. Web Appendix Table W.4 reports the sample accounting for the same-county and portfolio-wide (any-firm-quarter) restrictions.

Expiration creates a deadline, but peers still decide whether to exercise their leases. Favorable local information could both increase peers' exercise of expiring leases and affect nearby firms' adoption or investment. If recent information drives exercise, expiration-driven drilling should coincide with new leasing, wells drilled near expiration should be more productive where more expiring leases are exercised, or recent local productivity growth should precede more expiration-driven drilling. Internet Appendix Table OA.15 tests each implication. Expiration-driven drilling is uncorrelated with new leasing. Wells drilled in the expiration window show no productivity advantage where a larger share of expiring leases are exercised. Among counties with scheduled expirations, a one-standard-deviation increase in recent productivity predicts 0.007 fewer excess wells and a 0.9 percentage point lower probability of any expiration-driven drilling. The analyses cannot establish that lease exercise is random, but they provide no evidence that favorable contemporaneous information generates the expiration-related variation in peer activity.

5.2 Robustness of the Calendar and Vintage Patterns

The dynamic evidence makes two comparisons. The calendar results compare responses at different dates, so changes in oil prices or other aggregate conditions may explain the reversal. The block-vintage results compare formations that reach the same local-development stage at different dates, so fixed differences between early- and late-vintage formations may explain that contrast. I examine each possibility in turn.

Calendar-quarter fixed effects absorb the common oil-price level, but prices could still change firms' sensitivity to peer activity or network signals. Internet Appendix Table OA.16 adds oil-price controls and interacts prices with peer activity and each signal. The main extensive-adoption interactions remain close to the pooled estimates, while the triple inter-

actions with oil prices are imprecise. Because the same formation can produce both oil and gas, exposure to oil-price movements varies with its production mix. Internet Appendix Table OA.17 compares formations' pre-2014 gas shares with peer performance and technique intensity across periods. Both signals are more strongly associated with oil-heavy formations in 2008–14 and 2015–17, even though the adoption response changes sign between those periods. The correlations are near zero from 2018 onward. No result links oil-price exposure to the calendar reversal.

The block-vintage comparison could instead reflect fixed differences between formations that develop early and late. Internet Appendix Table OA.18 compares their first blocks. Late-vintage first blocks have lower average technique intensity, lower peer performance and lower well productivity than early-vintage first blocks. Their stronger adoption response is not accompanied by stronger measured signals or greater average productivity.

Fixed differences across vintages cannot explain changes within a vintage. Among early-vintage formations, the technique-intensity response in later blocks rises from -0.18^{**} through 2014 to 0.19^{***} in 2015–16. A direct test estimates the increase at 0.37^{***}. The estimate remains positive at 0.16 from 2017 onward. Local development also advances over these periods, so the comparison cannot isolate calendar timing. The within-vintage estimates make fixed differences across formation vintages an incomplete explanation for the dynamic pattern.

5.3 Alternative Explanations and Placebos

A general local productivity or demand boom could increase drilling without operating through the horizontal network signals. Internet Appendix Table OA.19 replaces the horizontal outcomes with vertical drilling. A general drilling boom should produce a parallel positive response in vertical activity. No parallel pattern appears. Peer performance is negatively associated with positive vertical drilling and vertical well counts, while technique

intensity and breadth produce no consistent positive response. Negative vertical estimates could reflect substitution toward horizontal drilling, so the placebo weighs against a general drilling boom without identifying the mechanism behind the horizontal response.

The interaction estimates alone provide an incomplete account of how firms respond to peer activity and network signals. Internet Appendix Table OA.20 reports the lower-order peer-activity and signal coefficients from the baseline specifications. For extensive adoption, the peer-performance and technique-intensity interactions are positive and precise, while their lower-order signal coefficients are smaller and imprecise. The peer-activity coefficient evaluates the marginal response at a signal value of zero, which lies outside these positive-signal estimation samples. It is not an estimate of the peer-activity response where no signal is measurable.

Internet Appendix Table OA.21 estimates that comparison directly using observations where none of the four network signals is distinguishable from zero. Peer activity has positive extensive responses of 0.0113^{***} for adoption and 0.0108^{**} for investment. The intensive-adoption response is also positive at the 10 percent level, while the intensive-investment response is imprecise. The same table shows that the no-signal group contains 2.7 percent of the panel and about 6 percent of observed adoption and investment. Network signals are not the only reason firms respond to peer activity, but most aggregate activity occurs where at least one signal is measurable. The small size of the no-signal group is itself informative about the prevalence of measured network structure.

Competition may also contribute to the estimated response. Peer drilling can provide information while increasing demand for rigs, crews, acreage and other local inputs. The estimates capture the net effect of these forces. A competition-only explanation would also need to account for the calendar reversal, the first-block difference across formation vintages and the opposing responses associated with technique intensity and breadth. Competition remains consistent with these patterns only if its effect varies systematically with timing and measured network content.

The firms that respond more strongly to local network signals change over the industry transition. Internet Appendix Figure OA.4 compares large and small firms and firms with more and less prior horizontal-production experience. In 2008–14, the high-minus-low response gaps are negative for peer performance and technique intensity. From 2015 onward, the gaps are positive. The reversal appears in both extensive adoption and extensive investment. Smaller and less experienced firms respond relatively more early. The relative response shifts toward larger and more experienced firms later.

The pattern shows that the capital-allocation response to the same local information depends on firm scale, prior experience and the stage of industry development. Early in the transition, peer activity may substitute for firms' limited experience with the technology. Later, firms with greater scale and experience may be better positioned to act on local information. The evidence does not identify either mechanism. Internet Appendix Table OA.22 finds no single ordering across finer categories of prior experience. The conceptual framework abstracts from this source of heterogeneity.

5.4 Measurement of Peer Activity and Network Signals

The main sign patterns remain robust to alternative variable constructions and data choices. The expiration spike remains when the bunching sample uses all spatially matched wells rather than the 0.25-mile cap, and when the estimation window widens, as reported in Internet Appendix Table OA.2. Separately, I vary how wells are assigned to leases. The baseline assigns each well to the nearest eligible lease within 0.25 miles. The alternative instead links wells to leases using standardized operator and lessee names, supplemented by a manual bridge between related entities. Internet Appendix Table OA.3 reports both the sharper expiration spike and the pooled response estimates under this construction. Other specifications vary which wells enter peer activity and when their effects appear. Restricting the measure to the first well on each lease removes 58 percent of bunching-

weighted activity but preserves the pooled signs (Internet Appendix Table OA.23). The estimates are strongest under tighter well-to-lease links, as shown in Internet Appendix Table OA.28. Jointly including four quarterly lags places the technique-intensity response in the baseline one-quarter lag, while peer performance is more persistent (Internet Appendix Table OA.11). Web Appendix Table W.6 reports the full four-signal grid. Dating peer activity at completion instead of spud preserves the signs but reduces precision (Internet Appendix Table OA.24).

6 Conceptual Framework

The reduced-form results show that firms' adoption response to local network information depends on the stage of the industry-wide transition and accumulated local development. They also separate the decision to adopt a production technology from the scale of investment conditional on using it. The two decisions respond differently early in the transition. This difference points to a role for adoption costs separate from the return to investment. I use a simple firm problem to organize these channels and a calibrated counterfactual to evaluate which combinations can reproduce the empirical patterns. The exercise is diagnostic rather than a structural estimate.

6.1 Model and Calibration

A firm chooses whether to adopt a new technology in a local market and, conditional on adoption, how much to invest. I normalize peer activity to one and write the firm's payoff from adoption as

$$M(S, n, N)d - \frac{\eta}{2}d^2 - F(S, n, N), \tag{3}$$

where S is the local signal, n is local development, and N is industry maturity. The function M is the operating return from the technology and F is the fixed cost of adoption.[23] The parameter η governs how quickly marginal investment costs rise. I hold η fixed. Appendix E.3.3 relaxes this, letting local information change the cost of scaling investment instead of the cost of starting. Firms differ in an adoption hurdle with distribution Ψ and density ψ. Optimal investment is $d^* = M/\eta$, adoption surplus is $\Pi = M^2/(2\eta) - F$, and the adoption share is $p = \Psi(\Pi)$. The responses to the local signal are

$$R^d(n, N) = \frac{M_S(n, N)}{\eta}, \qquad R^p(n, N) = \psi(\Pi) \left[\frac{M}{\eta} M_S(n, N) - F_S(n, N) \right]. \tag{4}$$

The return response M_S affects both decisions, while the starting-cost response F_S affects adoption without directly changing optimal investment. For the quantitative exercise, I impose $M_S(n, N) = (m_0 + m_1 N)e^{-bn}$ and $F_S(n, N) = f(1 - N)e^{-dn}$. In this exercise, N is measured by the industry-wide horizontal share, while $n = 0$ denotes a first block and $n = 1$ denotes a later block. After rescaling the investment estimates, I normalize η, M/η, and $\psi(\Pi)$ to one. The responses become

$$r^d(n, N) = (m_0 + m_1 N)e^{-bn}, \qquad r^p(n, N) = (m_0 + m_1 N)e^{-bn} - f(1 - N)e^{-dn}. \tag{5}$$

I use technique intensity for the benchmark calibration because its vintage-block estimates provide the adoption-investment comparisons needed to distinguish the return and starting-cost channels. Peer performance follows a different late-vintage pattern that may combine information about local returns with competition. Internet Appendix Section OA.5 separately calibrates its adoption responses. I fit the five parameters to ten technique-intensity estimates for adoption and investment levels conditional on positive investment across five mutually exclusive vintage-block groups. Appendix E derives the responses and Appendix E.3 reports

[23] Fixed costs and costly reversibility can separate the decision to use a technology from subsequent adjustments in investment scale (Abel and Eberly, 1996; Bazdresch, 2013).

the calibration.

6.2 Counterfactual Results

The quantitative exercise builds on two features of the vintage-block results. First-block responses differ according to when a formation develops in the industry-wide transition. Within the same formations, the response also changes as local development accumulates. A single set of five parameters must jointly reproduce all of these empirical patterns for adoption and investment across formation vintages and stages of local development. Table 4 reports the ten targets, their fitted values, and recalibrations that remove one model component at a time.

The full model fits the adoption responses more closely than the investment responses. It reproduces the negative adoption and positive investment responses in the early-vintage first block, the more positive first-block response in the late vintage, and the different first-to-later-block changes across vintages. The model misses the negative investment estimate for early-vintage later blocks from 2015. Panel B shows why the joint mechanism is needed. Setting $f = 0$ forces the adoption and investment responses to coincide. Setting $b = 0$ misses the positive late-vintage first-block response and its subsequent attenuation. Setting $d = 0$ reverses the sign of b and still misses the late-vintage ordering.

To interpret the full calibration, I evaluate the model at four combinations of industry maturity and local development. These calculations hold one state fixed as the other changes. They are not additional empirical estimates. Holding local development at the first-block state, moving from early- to late-vintage industry maturity raises the predicted adoption response from -0.42 to 0.40. Holding industry maturity at its early-vintage value, moving from the first to a later block raises the response from -0.42 to -0.07. In this comparison, the reduction in F_S exceeds the reduction in M_S. At late-vintage industry maturity, the same change in local development lowers the response from 0.40 to 0.12. The reduction in M_S

Table 4: Calibration fit and restricted models

Panel A. Empirical targets and full-model fit

	Adoption		Scaled investment	
Target	Estimate	Fitted	Estimate	Fitted
Early vintage, first block	-0.4597	-0.4248	1.1086	1.7681
Early vintage, later blocks through 2014	-0.1836	-0.1731	0.6396	0.4396
Early vintage, later blocks from 2015	0.2221	0.1711	-0.7177	0.3734
Late vintage, first block	0.8261	0.4028	0.8381	1.6219
Late vintage, later blocks from 2017	0.2668	0.2013	0.0453	0.3676

Panel B. Restricted recalibrations

Model	m_0	m_1	b	f	d	Loss	Main implication
Full model	2.73	-1.29	1.44	8.62	1.51	11.91	Matches the three qualitative comparisons in the notes
No starting-cost response, $f = 0$	-8.97	11.51	2.27	0	—	21.52	Adoption and investment responses coincide
No return attenuation, $b = 0$	0.89	-0.56	0	3.38	0.47	13.55	Misses the positive late-vintage first-block response and subsequent decline
No local decline in starting-cost response, $d = 0$	0.49	-0.33	-0.70	2.19	0	14.24	Requires returns to rise with local development and misses the late-vintage ordering

Notes. Panel A compares the model's fitted responses with ten technique-intensity targets. Adoption is the extensive-adoption response. Scaled investment is the response of horizontal well counts conditional on positive horizontal investment multiplied by 0.018588. Panel B reports the full calibration and three restricted models. The responses follow equation (5). m_0 is the return response to the signal in a young industry and m_1 its change with industry maturity. b governs how quickly the return response attenuates with local development. f is the starting-cost response in a young industry and d governs its decline with local development. The three qualitative comparisons are negative adoption alongside positive investment in early-vintage formations, a higher first-block adoption response in late-vintage than in early-vintage formations, and a decline in the late-vintage adoption response from the first block to later blocks. The early-vintage later-block target from 2015 combines the finer calendar groups reported in Table 3. Each restricted model is recalibrated to the same ten targets. The loss is the inverse-variance-weighted sum of squared deviations between the empirical and fitted responses. Appendix E.3 reports the calibration procedure.

then exceeds the reduction in F_S. The investment response exceeds the adoption response by $F_S = f(1-N)e^{-dn}$, which allows investment to remain positive when the adoption response is negative.

Figure 7 plots the full calibrated responses over the observed range of industry maturity. The curves hold local development fixed at either a first or later block. Movement along a curve reflects industry-wide development rather than a formation's history. Internet Appendix Figure OA.5 adds the model components sequentially using the same fitted parameters. Internet Appendix Figure OA.6 instead recalibrates the model after removing one component at a time.

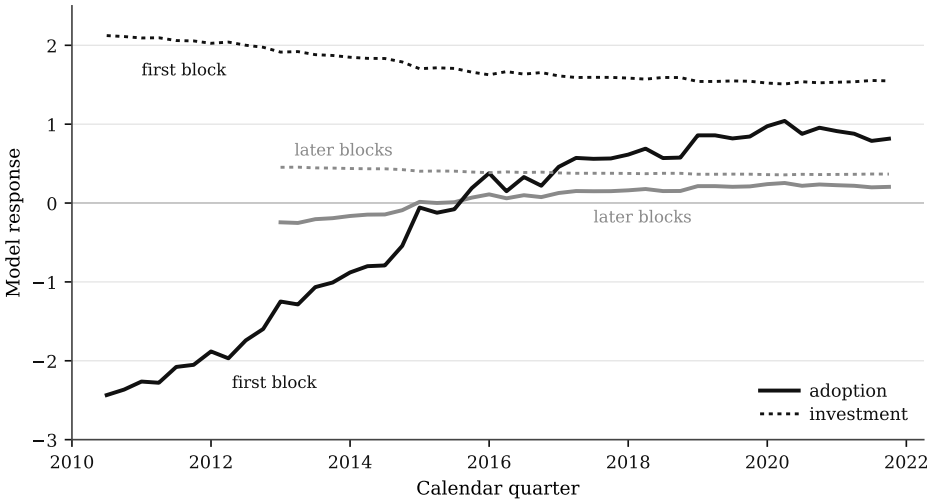


Figure 7: Full calibrated responses over the industry-wide transition

Notes. The figure evaluates equation (5) at the industry-wide horizontal share observed each quarter. Black curves hold local development at a formation's first 2,000-well block, $n = 0$. Gray curves hold it at subsequent blocks, $n = 1$. Solid lines report normalized adoption responses. Dotted lines report normalized investment responses. The curves hold local development fixed and are not formation histories. The horizontal axis, labeled "Calendar quarter," is the quarter whose industry-wide horizontal share enters the model. The vertical axis, labeled "Model response," reports the normalized responses.

The calendar estimates in Figure 6 average across formations at different stages of local development. They are not among the ten calibration targets. I construct their model counterpart by evaluating the fitted response for each observation using its quarter's industry maturity and local-development stage, then averaging within the same three periods. The estimated responses are −0.17, 0.36, and 0.28. The corresponding model predictions are −0.70, 0.20, and 0.24. The model reproduces the negative-to-positive reversal but not the subsequent flattening. It overstates the initial negative response and predicts a slight increase rather than the observed decline after 2017. The comparison shows that the mechanisms fitted to the vintage-block estimates generate the broad calendar reversal after accounting for the changing composition of local-development stages.

Within this model, reproducing the empirical patterns requires three forces to operate jointly. First, when the industry and local market are young, a stronger local signal must raise adoption-specific costs enough to offset its return value for potential adopters even as

firms already using the technology increase investment. The parameter $f > 0$ represents this initial cost sensitivity. Economically, it may arise when firms responding to the same information compete for scarce specialized inputs needed to adopt. Second, the net adoption response in a first block must become more positive as the industry develops. This requires $m_1 + f > 0$, so the change can reflect an increase in the return value of local information, a reduction in its effect on adoption costs, or both. In the fitted calibration, $m_1 < 0$, so the vintage difference comes from the cost effect declining by more than the return effect. Third, $b > 0$ makes the return value of additional local information attenuate as local experience accumulates at any level of industry maturity. This does not imply that the net adoption response always declines with local development. Early in the industry-wide transition, the cost effect remains large and its attenuation through $d > 0$ dominates. Later, the cost effect is smaller and return attenuation produces the decline across local-development blocks.

7 Conclusion

The relationship between information propagated through local networks and technology adoption changes over an industry-wide transition. I establish this result by combining local network signals measured from earlier wells with lease-expiration-induced peer drilling. Peer drilling makes new local activity observable, although the estimates capture its net informational and competitive effects, not a pure information channel. The vintage-block estimates provide the clearest evidence. In first blocks, the late-minus-early adoption difference is 0.38^{***} for peer performance and 1.29^{**} for technique intensity. Because these comparisons describe the same stage of local development, accumulated local experience cannot fully explain them. After both vintages reach later blocks, technique-intensity responses converge toward zero while peer performance develops a significant negative late-minus-early difference. Local development and industry timing both contribute to the adoption response, but the two signals also contain different economic information.

The difference between adoption and investment helps distinguish the economic mechanisms consistent with this pattern. In early-vintage first blocks, technique intensity has a negative adoption response and a positive but imprecise investment-level response. Within the model, this contrast supports an adoption cost separate from the return governing investment, consistent with competition for scarce specialized resources. The calibration combines an adoption-cost effect that declines with industry and local development with diminishing returns to local information. Removing each component and refitting the remaining parameters causes the model to miss part of the vintage-block pattern. The same fitted parameters also generate the broad calendar reversal. Economically, industry development reduces the cost effect by more than the return benefit, making the adoption response more positive. Local experience weakens both effects, so its consequences for adoption depend on the stage of industry development.

The findings imply that the effect of information on capital allocation depends on the broader technology transition. Industry-wide development changes the value or cost of acting on local evidence. Accumulated local development determines how much an additional signal contributes. The results also caution against equating more disclosure with more useful information. Peer-technique breadth has no positive adoption response even though technique intensity, constructed from the same disclosed attributes, does. Policies that expand information may have limited effects when firms lack the capacity, specialized inputs, or standardized practices needed to act on it. The evidence cannot determine which constraint dominates, so it supports this complementarity without selecting a particular intervention.

The firm-heterogeneity results suggest an important direction for future research. Relative responsiveness shifts from smaller and less experienced firms early in the transition toward larger and more experienced firms later. The shift in responsive firm types could contribute to the aggregate timing pattern or accompany changes in the value of local evidence. Further analysis could examine firm type and adoption capacity as well as the reverse relationship through which successful local clusters contribute to industry-wide development.

The analysis here establishes that the stage of the broader transition conditions what local networks contribute to firm adoption and investment.

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A Peer-Activity and Lease-Link Construction

Section 3.1 describes the baseline spatial assignment. Within each county, I match a well to leases signed before its spud date and located within 0.25 miles. I assign the nearest lease. When multiple leases lie within 0.01 miles of the nearest lease, I choose the earliest expiration date, then the earliest signing date, then the smallest lease identifier. The spatial assignment does not require the standardized operator and lessee names to agree. A first linked well is the earliest well assigned to a lease and is used only in Figure 1. The bunching estimator uses all uniquely assigned wells.

The alternative construction requires the standardized operator name to match the leaseholder name after applying the manual name bridge. I join operator wells to leases held by the matched lessee within the same county, retaining leases signed before the well was spudded. When a well matches multiple leases, I assign the earliest expiration date, then the earliest signing date, then the smallest lease identifier. The event-time estimator is otherwise unchanged. Internet Appendix Table OA.3 reports the resulting response estimates.

For each assigned well w, event time is $\tau_w = q_w^{spud} - q_w^{expire}$, where q_w^{spud} is the spud quarter and q_w^{expire} is the expiration quarter of the assigned lease. Let C_τ be the national count of assigned wells at event time τ, using the main spatial match. I estimate the bunching regression

$$C_\tau = \alpha + \sum_{p=1}^{5} \eta_p \tau^p + \sum_{k \in \mathcal{B}} \lambda_k \mathbf{1}\{\tau = k\} + u_\tau, \tag{A.1}$$

where $\mathcal{B} = \{-4, -3, -2, -1, 0, 1\}$ is the prespecified expiration window. The fifth-order polynomial is estimated over $\tau \in [-12, 4]$. Setting the bunching-window indicators to zero gives the smooth counterfactual count $\widehat{C}_\tau^0$. The estimated excess share at each event time is

$$\widehat{\pi}_\tau = \mathbf{1}\{\tau \in \mathcal{B}\} \max\left\{0, \min\left\{1, \frac{C_\tau - \widehat{C}_\tau^0}{C_\tau}\right\}\right\}, \tag{A.2}$$

with the share set to zero when $C_\tau = 0$. An assigned well drilled near expiration receives fractional weight $\widehat{\pi}_{\tau_w}$ from equation (A.2), the estimated share of wells at that event time above the smooth counterfactual.

The excess mass remains positive under third-, fifth-, and seventh-order polynomials. The seventh-order estimates are larger, while the ninth-order specification leaves only one residual degree of freedom and is uninformative. I retain the fifth-order specification as the baseline. Internet Appendix Section OA.2.1 reports the estimates and standard errors.

The county-quarter excess peer-activity measure sums these fractional weights among wells spudded in county c during quarter q, $\widehat{E}_{cq} = \sum_{w \in (c,q)} \widehat{\pi}_{\tau_w}$. For firm i, I remove the firm's own operated and leased wells from this county-quarter total. With operated and leased components denoted by $\widehat{E}^{op}_{icq}$ and $\widehat{E}^{les}_{icq}$, and with $\widehat{E}^{both}_{icq}$ restoring wells for which the firm is both operator and lessee, the firm-excluded measure is $\widehat{E}^{-i}_{cq} = \widehat{E}_{cq} - \widehat{E}^{op}_{icq} - \widehat{E}^{les}_{icq} + \widehat{E}^{both}_{icq}$. The baseline response specifications use $\widehat{E}^{-i}_{c,q-1}$, the one-quarter lag. Internet Appendix Table OA.11 reports the corresponding specification with lags one through four entered jointly.

B Productivity and Skill Measurement

I construct the productivity and skill inputs to the network regressions using a modified decline-curve approach built around the classic Arps production model. Because well output is highly age-dependent, raw monthly production conflates underlying productivity with well age. Arps-style decline curves impose a parsimonious parametric shape on the production path as a function of well age. In its common hyperbolic form,

$$q(t) \;=\; \frac{q_0}{(1 + bD_0 t)^{1/b}},$$

where q_0 is an initial production level, D_0 is a decline parameter, and b governs curvature.

First, I estimate a common hyperbolic decline curve $\widehat{q}(\cdot)$ as a function of age and form an

age-adjusted production residual, $\log(q_{it}/\widehat{q}(age_{it}))$. From this residual I construct a productivity index, comparable across wells, that serves as the dependent variable Y in the spatial regressions. The decline curve is fitted once on all well-months in the production data, for wells that began producing in 2005 or later, through the end of the data in 2022. A well's productivity is its average residual over every month in which it reports production. For a well in a given block, that average includes production reported after the block closes. The network signals classify each block using the full subsequent production record. They are not limited to information available when the block closed.

Second, I use the same decline adjustment to construct a time-varying firm skill proxy. Using the pooled sample, I regress decline-adjusted production on location controls and absorb firm-quarter fixed effects. The estimated firm-quarter component captures systematic differences in performance across firms within a quarter, net of the common decline structure and observed location. I use this component as the firm-quarter skill proxy Z that enters the spatial regressions through WZ. Because this proxy is estimated from the firm's broader operated portfolio across formations, not only wells in the firm's own formation, it captures firm-level performance differences beyond purely local outcomes.

C Spatial Network Model, Weights, and Blocks

The baseline spatial model links a well's productivity to the performance and skill of earlier wells drilled by other firms in the same formation. All models are estimated separately by formation and, in the dynamic construction, separately within each completed 2,000-well block. For well i in formation g, the model is $Y_i = \gamma_1^g \sum_{j \neq i} w_{ij} Y_j + X_i' \gamma_2^g + \beta^g \sum_{j \neq i} w_{ij} Z_j + \nu_i$, or in matrix notation, $\mathbf{Y} = \gamma_1^g \mathbf{WY} + \mathbf{X}\gamma_2^g + \beta^g \mathbf{WZ} + \nu$. Here $\mathbf{Y}$ is well productivity, $\mathbf{Z}$ is firm skill and $\mathbf{W}$ defines peer exposure. The element w_{ij} is nonzero only when peer well j was drilled earlier by a different firm in the same formation and lies inside the relevant local neighborhood. The products $\mathbf{WY}$ and $\mathbf{WZ}$ are weighted peer performance and weighted

peer skill. The coefficient γ_1^g is the peer-performance signal, WY, and β^g is the peer-skill signal, WZ. The control vector X_i contains five firm controls and the same five measures for the well's county. They are the horizontal share of production, the horizontal well count, the total well count, the standard deviation of perforated-interval length among wells that began producing in the same year, and the mean first-year output of those wells.

I estimate each network object under four weighting schemes. Let t_i be well i's first production month, d_{ij} the great-circle distance in miles between wells i and j, $o(i)$ the operating firm and z_i the skill proxy for the operator of well i. The weights are directional. A peer well can enter row i only if it is in the same formation, began producing earlier and was drilled by another firm. Peers are ordered by first production month, not by drilling date. For the distance-based, distance-time and relative-skill weights, the eligible peer set is $\mathcal{J}_i = \{j : g_j = g_i,\ o(j) \neq o(i),\ 0 < t_i - t_j \leq 36, d_{ij} \leq 50\}$. The unnormalized inverse-distance entries are $\widetilde{w}_{ij}^{ID} = \mathbf{1}\{j \in \mathcal{J}_i\}/d_{ij}$. Distance-time entries are $\widetilde{w}_{ij}^{DT} = \mathbf{1}\{j \in \mathcal{J}_i\}/[d_{ij}(t_i - t_j)]$. Equidistant common-period entries are $\widetilde{w}_{ij}^{EQ} = \mathbf{1}\{j \in \mathcal{J}_i\}\mathbf{1}\{t_i - t_j \leq 4\}$. Relative-skill entries are $\widetilde{w}_{ij}^{RS} = \mathbf{1}\{j \in \mathcal{J}_i\} \max\{z_j - z_i, 0\}$. The estimation uses these entries without row normalization. Inverse-distance weights place more weight on nearby prior wells. Distance-time weights multiply inverse distance by inverse elapsed time, so the most influential peers are both nearby and recent. Equidistant common-period weights give equal weight to prior peer wells in a short local development episode. Relative-skill weights place weight on earlier wells drilled by more-skilled firms, with weight increasing in the skill gap between the peer firm and the operator of the current well.

Within each formation, I sort horizontal wells by first production month and divide them into consecutive, non-overlapping blocks of 2,000 wells. I drop the final incomplete block and estimate each completed block separately. A block's regressions use only wells within that block, so its peer set does not include wells from earlier blocks. I date each block by its first and last production quarters. A response observation receives the signal from the most recently completed block in its formation.

Table D.1: Peer-technique attributes used to construct breadth and intensity

Category	Attribute
Hydraulic-fracturing input	Proppant
Hydraulic-fracturing input	Biocide
Hydraulic-fracturing input	Breaker
Hydraulic-fracturing input	Buffer
Hydraulic-fracturing input	Clay control
Hydraulic-fracturing input	Crosslinker
Hydraulic-fracturing input	Friction reducer
Hydraulic-fracturing input	Gelling agent
Hydraulic-fracturing input	Iron control
Hydraulic-fracturing input	Scale inhibitor
Hydraulic-fracturing input	Surfactant
Hydraulic-fracturing input	Energizer
Hydraulic-fracturing input	Diverter
Pressure reading	Casing pressure
Pressure reading	Flowing tubing pressure
Pressure reading	Shut-in pressure

Notes. Each listed attribute is converted to an availability indicator equal to one when the underlying well record reports that field. The peer-technique models use weighted peer exposure to these availability indicators, not the reported quantities themselves.

D Peer-Technique Measurement

Let $k \in \mathcal{K}$ index one of the $K = 16$ reported design and operating attributes listed in Table D.1.

For each attribute, I estimate a separate spatial regression that replaces the peer-skill term in the baseline spatial model with exposure to peers whose well records report attribute k:

$$\mathbf{Y} = \gamma_1^g \mathbf{WY} + \mathbf{X}\gamma_2^g + \beta_k^g \mathbf{WA}^k + \nu. \tag{D.1}$$

$\mathbf{A}^k$ is an $N \times 1$ indicator vector, where $A_i^k = 1$ if attribute k is reported for well i. The coefficient β_k^g measures the association between productivity and exposure to peers whose records report attribute k, conditional on the baseline spatial lag WY and controls.

Breadth is the count of attributes whose peer-exposure coefficient is positive and whose 95% confidence interval excludes zero. To construct technique intensity, I include all K peer-technique exposure terms jointly in the same model. Intensity is the partial R^2 from adding those terms to the baseline model with WY and controls. The corresponding Wald test evaluates whether the peer-technique terms are jointly significant.

E Conceptual Framework Derivations and Calibration

E.1 Firm Problem

Peer activity is normalized to one. Conditional on adoption, the firm solves the problem below. The parameter $\eta > 0$ is independent of the local signal.

$$\max_{d \geq 0} \left\{ M(S, n, N)d - \frac{\eta}{2} d^2 - F(S, n, N) \right\}.$$

For an interior solution, optimal investment conditional on adoption and the surplus from adoption are

$$d^*(S, n, N) = \frac{M(S, n, N)}{\eta},$$

$$\Pi(S, n, N) = \frac{M(S, n, N)^2}{2\eta} - F(S, n, N).$$

Firm i adopts when $\Pi(S, n, N) \geq \xi_i$. If Ψ is the distribution of ξ_i, the adoption share is $p(S, n, N) = \Psi(\Pi(S, n, N))$. The investment response to the local signal is

$$\frac{\partial d^*}{\partial S} = \frac{1}{\eta} M_S. \tag{E.1}$$

The corresponding adoption response is

$$\frac{\partial p}{\partial S} = \psi(\Pi)\left[\frac{M}{\eta}M_S - F_S\right]. \tag{E.2}$$

E.2 Parameter Restrictions

Under the benchmark normalizations, the predicted investment and adoption responses are

$$r^d(n,N) = (m_0 + m_1 N)e^{-bn}, \qquad r^p(n,N) = (m_0 + m_1 N)e^{-bn} - f(1-N)e^{-dn}.$$

Let N_{EF}, N_{EL}, N_{LF}, and N_{LL} denote industry maturity in the early-vintage first block, early-vintage later blocks through 2014, late-vintage first block and late-vintage later blocks from 2017. Their measured values are 0.745, 0.678, 0.858 and 0.913.

In the early-vintage first block, the adoption estimate is negative and the intensive-investment point estimate is positive. The model counterpart is

$$r^d(0,N_{\mathrm{EF}}) = m_0 + m_1 N_{\mathrm{EF}} > 0, \qquad r^p(0,N_{\mathrm{EF}}) = m_0 + m_1 N_{\mathrm{EF}} - f(1-N_{\mathrm{EF}}) < 0.$$

Together these imply $f(1-N_{\mathrm{EF}}) > m_0 + m_1 N_{\mathrm{EF}} > 0$. Because $N_{\mathrm{EF}} < 1$, the adoption-investment contrast requires $f > 0$.

The first-block vintage comparison subtracts two responses at $n = 0$. Thus

$$r^p(0,N_{\mathrm{LF}}) - r^p(0,N_{\mathrm{EF}}) = (m_1 + f)(N_{\mathrm{LF}} - N_{\mathrm{EF}}).$$

Because $N_{\mathrm{LF}} > N_{\mathrm{EF}}$, a higher late-vintage first-block response requires $m_1 + f > 0$.

For late-vintage blocks, the observed comparison is

$$(m_0 + m_1 N_{\mathrm{LL}})e^{-b} - f(1-N_{\mathrm{LL}})e^{-d} < m_0 + m_1 N_{\mathrm{LF}} - f(1-N_{\mathrm{LF}}).$$

Given $N_{\mathrm{LL}} > N_{\mathrm{LF}}$, $m_1 + f > 0$ and $d \geq 0$, this inequality cannot hold when $b = 0$. Matching the point-estimate ordering requires enough saturation, $b > 0$, to offset the increase in industry maturity. The direct first-to-later difference is imprecise, so the ordering does not identify b separately.

For early-vintage blocks, the comparison is

$$r^p(1, N_{\mathrm{EL}}) - r^p(0, N_{\mathrm{EF}}) = \Big[(m_0 + m_1 N_{\mathrm{EL}})e^{-b} - (m_0 + m_1 N_{\mathrm{EF}})\Big] - f\Big[(1 - N_{\mathrm{EL}})e^{-d} - (1 - N_{\mathrm{EF}})\Big].$$

If $d = 0$, $b \geq 0$ and $m_1 \geq 0$, the first bracket is not positive and the second equals $N_{\mathrm{EF}} - N_{\mathrm{EL}} > 0$. The later-block response would be lower. Under these sign restrictions, matching the observed ordering requires $d > 0$. This is a sufficient result rather than general identification of d. The unrestricted calibration permits $m_1 < 0$. Its $d = 0$ fit preserves the early-vintage ordering only by setting $b < 0$, which conflicts with the late-vintage decline and calendar trajectory.

The calendar response in period t averages the model response over that period's estimation sample $\mathcal{S}_t$:

$$\bar{p}_t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} r^p\big(n_i, N_{q(i)}\big).$$

Because this expression depends on each period's mix of first and later blocks, the calendar pattern is evaluated numerically rather than imposed as a separate parameter restriction.

E.3 Calibration Details

Table E.1 reports the extensive-adoption and intensive-investment targets for five mutually exclusive technique-intensity groups. Let $\widehat{r}_k^{\,p}$ and $\widehat{r}_k^{\,d}$ denote the two responses for group k. I set $n_k = 0$ for first blocks and $n_k = 1$ for later blocks. I measure N_k with the industry-wide horizontal share averaged over the observations in each group's estimation sample, which weights quarters as in the regression.

Table E.1: Calibration targets

Target	n_k	N_k	Adoption	Investment	Scaled investment
Early vintage, first block	0	0.7455	-0.4597^{**}	59.64	1.1086
			(0.2301)	(74.62)	(1.3870)
Early vintage, later blocks through 2014	1	0.6777	-0.1836^{**}	34.41^{***}	0.6396
			(0.0740)	(11.57)	(0.2150)
Early vintage, later blocks from 2015	1	0.8936	0.2221^{*}	-38.61^{**}	-0.7177
			(0.1242)	(18.78)	(0.3491)
Late vintage, first block	0	0.8585	0.8261	45.09	0.8381
			(0.5978)	(156.20)	(2.9035)
Late vintage, later blocks from 2017	1	0.9125	0.2668^{*}	2.44	0.0453
			(0.1462)	(68.10)	(1.2658)

Notes. The targets are technique-intensity extensive-adoption and intensive-investment estimates for five mutually exclusive formation-vintage, local-block, and calendar-period groups. The early-vintage later-block target from 2015 combines the finer calendar groups reported in Table 3. Local development is normalized to $n_k = 0$ for first blocks and $n_k = 1$ for later blocks. Industry maturity N_k is the industry-wide horizontal share averaged over the adoption estimation sample for target k. Scaled investment multiplies the intensive-investment estimate and its standard error by 0.018588. The calibration weights each target by the inverse of its reported formation-block-clustered sampling variance. Standard errors, clustered by formation-block as used in the calibration, are in parentheses. They come from the per-vintage regressions and differ slightly from Table 3, which estimates all cells in one pooled regression. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

I impose the scale normalizations $\psi(\Pi) = 1$, $M/\eta = 1$ and $\eta = 1$. The resulting responses are given in equation (5). The parameter vector $\theta = (m_0, m_1, b, f, d)$ minimizes

$$\mathcal{L}(\theta) = \sum_{k=1}^{5} \left[\frac{(r_k^p(\theta) - \widehat{r}_k^p)^2}{(\mathrm{se}_k^p)^2} + \frac{\left(r_k^d(\theta) - \widehat{r}_k^d\right)^2}{(\mathrm{se}_k^d)^2} \right]. \tag{E.3}$$

Investment estimates and standard errors are multiplied by 0.018588, the slope from a regression through the origin of the two late-vintage adoption estimates on the corresponding intensive-investment estimates. The factor is fixed and each target is weighted by its inverse sampling variance. These choices set the relative scale of the two outcomes but do not change their signs.

The minimization places no sign restrictions on the parameters. Conditional on b and d, the responses are linear in m_0, m_1 and f, which are obtained by weighted least squares. I profile the loss over a two-dimensional grid for b and d, then polish the best grid points.

Because n takes only two values, the targets identify first-to-later-block ratios rather than the exponential shape between them.

The fitted values are $m_0 = 2.73$, $m_1 = -1.29$, $b = 1.44$, $f = 8.62$, and $d = 1.51$. Table 4 reports the fitted responses and the restricted recalibrations. The full-model loss is 11.9094. The adoption component is 0.9137 and the investment component is 10.9957.

Internet Appendix Figure OA.6 recalibrates the model after setting $f = 0$, $b = 0$, or $d = 0$. Removing f forces adoption and investment to coincide. Removing b or d reverses the fitted ordering of late-vintage first and subsequent blocks. Each restricted model is recalibrated to the same ten targets.

E.3.1 Calendar Prediction

The calendar responses in Figure 6 are not calibration targets. I calculate them by evaluating equation (5) for each observation in the corresponding period regression, using its block position and its quarter's industry-wide horizontal share, then averaging over the estimation sample. The predicted responses are -0.702, 0.202, and 0.236. The model reproduces the reversal from the first period to the second but not the subsequent flattening. Its predicted response rises slightly after 2017.

E.3.2 Why the Cost Depends on Local Development

Setting $d = 0$ fixes the starting-cost response across local development. The best fit has a loss of 14.2385 and sets $b = -0.6970$, so the benefit rises rather than saturates with local development. It predicts -0.11 for the late-vintage first block and 0.18 for its later blocks, with calendar responses of -0.35, -0.04 and 0.18. The restricted model preserves the early-vintage ordering only by reversing benefit saturation. It misses the positive late-vintage first-block response and the subsequent decline.

E.3.3 Relaxing the Constant-η Assumption

Section 6.1 assumes that η is constant. The alternative sets $F_S = 0$ and allows peer activity to change both the return to investment and the cost of scaling as a function of the local signal. Under the same normalizations,

$$r_k^d = M_{S,k} - \eta_{S,k}, \qquad r_k^p = M_{S,k} - \frac{1}{2}\eta_{S,k}.$$

I use

$$M_S(n, N) = (\alpha + \gamma N)e^{-\beta n}, \qquad \eta_S(n, N) = -q(1 - N)e^{-\delta n}, \tag{E.4}$$

with $\gamma, \beta, q, \delta > 0$. Early in the episode, $M_S < 0$ reduces adoption while $\eta_S < 0$ lowers the cost of scaling and raises optimal investment conditional on positive investment. The condition $M_{SN} > 0$ makes the return response less negative as industry maturity increases. Local development attenuates the return response toward zero. Consequently, $M_{Sn} > 0$ while $M_S < 0$ and $M_{Sn} < 0$ after the return response becomes positive.

The fitted values are $\alpha = -14.30$, $\gamma = 15.87$, $\beta = 1.51$, $q = 15.79$, and $\delta = 1.42$. A stronger signal lowers the return to investment early in the episode, $M_S < 0$, while industry maturity raises that return. The negative scaling-cost response produces positive investment, and the alternative reproduces the block trends. The drill-time evidence in Internet Appendix Table OA.10 cannot distinguish the two versions because drill time per well is not the curvature of cost in the number of wells. The technique-intensity estimate is positive at lag three but imprecise at the other lags, and the sample starts in 2014Q4. Setting $\gamma = 0$ also sets $M_{SN} = 0$ and removes the reversal.

Internet Appendix for Local Networks and Technology Adoption during Industry-Wide Technological Change

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OA.1 Overview

This Internet Appendix presents the discussion first and collects all figures and tables after the prose. Section OA.2 reports additional evidence on the empirical framework and identification. Section OA.3 provides supplemental results for the pooled, calendar and formation-vintage analyses. Section OA.4 organizes the remaining robustness evidence around expiration-driven drilling, dynamic patterns, alternative explanations and measurement. Section OA.5 reports additional illustrations for the conceptual framework. Additional specification results that do not bear on the discussion are reported for completeness in a Web Appendix available at `https://www.nverachau.com/assets/papers/tipping_empirical-web_appendix.pdf`.

OA.2 Empirical Framework and Identification

Figure OA.1 illustrates the technology distinction used to measure adoption. A horizontal well turns through the target formation and combines a lateral wellbore with staged hydraulic fracturing. A conventional vertical well accesses only the rock surrounding its vertical bore.

Figure OA.2 reports the distribution of the one-quarter-lagged, firm-excluded peer-activity measure across response samples and formations.

County-quarter drilling conditions could create apparent expiration bunching without a distinct lease deadline response. Table OA.2 adds county-by-quarter fixed effects to the event-time count model and varies the link radius and event window. The increase in drilling near expiration remains under these specifications.

OA.2.1 Polynomial Order for the Bunching Counterfactual

The baseline event-time specification in Appendix A of the main paper uses a fifth-order polynomial. I re-estimate the national excess counts with polynomials of order three, seven, and nine, holding the expiration window and the 0.25-mile sample fixed. The excess count in the expiration quarter

is 339.8 at order three, 322.8 at order five, and 510.7 at order seven, with standard errors of 75.4, 90.3, and 123.7. The excess count one quarter before expiration is 299.0, 305.7, and 552.5, with standard errors of 75.9, 102.7, and 146.0. The ninth-order specification has one residual degree of freedom. Its corresponding estimates are 2,994.7 and 3,380.2, with standard errors of 1,529.2 and 1,696.5. The ninth-order estimates are uninformative. The baseline weights and response estimates use the fifth-order specification.

The spatial lease assignment may incorrectly link wells to nearby leases. Table OA.3 instead links operators and lessees by name. Panel A reports an expiration-quarter excess share of 62.75 percent under this alternative construction, compared with 37.14 percent one quarter before expiration. Panel B rebuilds peer activity from the name-matched links and re-estimates the pooled response. The signs of the peer-performance, peer-technique breadth and technique-intensity estimates agree with the baseline, although the alternative measure has a different scale and produces less precise estimates.

Peer activity is estimated from the lease-expiration bunching curve. Table OA.4 accounts for this generated regressor by resampling counties, re-estimating the bunching curve and reconstructing peer activity in each draw. The percentile intervals preserve the principal extensive-adoption results and the technique-intensity result for intensive investment.

Web Appendix Table W.1 reports all four preferred network measures for each completed formation block.

Table OA.5 compares each alternative spatial weight with the preferred measure on the same completed formation blocks. The table reports correlations in signal values and agreement in the positive-signal classification used to define the response samples. Holding the formation block fixed separates sensitivity to spatial weights from changes in local development.

Signals are updated when a formation completes another 2,000-well block. Peer activity from the block-closing quarter could enter the first outcome quarter that uses the newly estimated signal. Table OA.6 drops that outcome quarter after every block closure. The principal pooled signs remain unchanged.

OA.3 Supplemental Empirical Results

The magnitude calculations in the introduction use the mean signal values in Table 1 of the main paper. Web Appendix Table W.3 reports the outcome, peer-activity and signal distributions within the peer-performance and technique-intensity response samples.

The network signals use different units. Their raw coefficients are not directly comparable. Table OA.7 standardizes peer activity and each signal within its estimation sample. The estimates describe a one-standard-deviation change in each component, although differences in the signal-specific samples still limit comparisons across signals.

Technique intensity is built from indicators for whether peers report each drilling and completion attribute. Disclosure rules vary across states and over time. The number of reported attributes changes across formation-blocks. Table OA.8 replaces technique intensity with the average number

of attributes reported per well in the formation-block. Reporting volume does not reproduce the baseline response in the full panel. Within the positive technique-intensity sample, reporting volume and intensity are too closely related to separate their effects.

Peer-technique breadth and technique intensity capture different features of reported technique information. Table OA.9 describes the attributes added when breadth increases and the extent to which those attributes overlap in what they explain. Table OA.10 reports whether the four network signals change the design and execution of firms' own wells.

The baseline estimates use peer activity from the preceding quarter. Table OA.11 enters lags one through four jointly to examine persistence for the two headline signals and outcomes. The technique-intensity response is concentrated at the first lag, while the peer-performance adoption response persists at longer lags. Web Appendix Table W.6 reports the complete four-signal, four-outcome grid.

The preferred period specification allows the controls and fixed effects to vary across eras. Table OA.12 compares it with a specification that holds both common across eras. The choice matters most for the early technique-intensity response, which is -0.174^{***} under the preferred specification and 0.228 and imprecise under common controls. Peer performance retains a positive middle-period estimate and a negative later estimate under both specifications.

The main period figure displays extensive adoption and intensive investment for peer performance and technique intensity. Table OA.13 reports intensive investment for all three signals, the two margins not plotted in the main paper and the period estimates for peer-technique breadth. It suppresses only the peer-performance and technique-intensity extensive-adoption estimates already shown in the main figure.

The main formation-vintage table reports the aligned extensive-adoption comparisons. Table OA.27 reports the corresponding aligned intensive-investment estimates, including the 2019 onward comparison.

OA.4 Robustness

OA.4.1 Endogeneity of Expiration-Driven Drilling

The baseline removes firm-county-quarters in which the responding firm has a same-county lease expiration. This restriction removes 0.31 percent of the full panel. A stricter restriction removes every observation for a firm-quarter in which the firm has an expiration anywhere in its portfolio, eliminating 13.3 percent of the panel. Table OA.14 re-estimates the pooled responses under that portfolio-wide restriction. It also allows the responding firm's own deadline pressure to extend beyond the expiration quarter and applies a full-window same-county exclusion, which removes 1.03 percent of the panel, to the pooled and vintage-block specifications. The principal pooled signs and first-block vintage contrasts remain similar. Web Appendix Table W.4 reports the sample accounting for the same-county and portfolio-wide (any-firm-quarter) restrictions.

A leaseholder facing a deadline chooses whether to drill or let the lease lapse. If that choice fa-

vored places with strong local returns, county-quarters with heavy expiration-driven drilling would also be places with good local conditions. Table OA.15 reports three analyses. Panel A tests whether expiration-driven drilling coincides with new lease acquisition in the same county. Panel B tests whether higher exercise rates select wells with greater first-year output. Panel C tests whether recent productivity improvements predict more expiration-driven drilling among counties with leases scheduled to expire. None of the three analyses indicates that favorable contemporaneous information generates greater expiration-driven drilling.

OA.4.2 Calendar and Formation-Vintage Patterns

The calendar reversal overlaps with large changes in oil prices. Table OA.16 allows prices to change the response to peer activity and network signals. The pooled signal interactions remain close to the baseline estimates. The triple interactions with oil prices are imprecise.

The oil-price analysis depends on how exposed each formation is to oil rather than gas prices. Table OA.17 reports the cross-formation correlation between each principal signal and the formation's pre-2014 gas share. Peer performance and technique intensity are more strongly associated with oil-heavy formations in both 2008–14 and 2015–17. The correlations are near zero from 2018 onward.

The block-vintage comparison could also reflect fixed differences across formations. Table OA.18 compares signal levels and well productivity in the first blocks of early- and late-vintage formations.

OA.4.2.1 Local Signal Trajectory Robustness

The trajectory exercise is a robustness check on whether the timing results are driven only by calendar period. Figure OA.3 uses network signal trajectories instead of formation vintage. For each formation and signal, I classify the path across successive 2,000-well blocks as decreasing, flat or increasing. I then estimate the peer-activity response separately for decreasing, flat and increasing paths, within early and late cohorts. The classification is done separately for each network signal, so a formation can be increasing for peer performance but flat for technique intensity.

Formations with decreasing peer-performance trajectories do not have falling productivity. In these formations, nearby peer productivity becomes less predictive of cross-sectional productivity differences as local development advances. Productivity growth is larger in these formations than in flat peer-performance formations in a block-level diagnostic.[1] The responses in these formations are not weaker than in flat paths. For extensive adoption, the decreasing-path estimate exceeds the flat-path estimate in both cohorts, at 0.93^{***} against 0.14^{**} in 2008–14 and 0.16 against -0.03 from 2015 onward. For intensive investment in 2008–14, it is 400.4^{***} against 14.8^{*}. From 2015 onward neither intensive-investment estimate is significant.

[1]The diagnostic groups formation blocks by the peer-performance trajectory classification and computes productivity growth relative to each formation's first block. Mean log-productivity growth is 0.448 in decreasing peer-performance paths and 0.351 in flat peer-performance paths. The analogous growth in the well fixed-effect component of productivity is 0.326 and 0.234. In blocks 2, 3 and 4, log-productivity growth is 0.511, 0.713 and 0.753 for decreasing peer-performance paths, compared with 0.279, 0.448 and 0.542 for flat peer-performance paths.

For technique-based signals, the trajectory split distinguishes broader disclosure from more informative implementation content. Increasing peer-technique breadth fails to generate stronger adoption or intensive-investment responses. More measured attributes are not enough. Technique intensity is different. The late-period adoption response is positive and significant in increasing-intensity paths, consistent with the cohort result. The opposite appears in decreasing-intensity paths. Where technique intensity declines across blocks, the late adoption response is strongly negative, at -9.07^{***}. The early intensive-investment response is also negative, at -197.3^{***}. I do not use the early increasing-intensity group as a clean contrast because that group has little support. The useful reading is narrower. Technique intensity matters for adoption when the implementation signal is still strengthening locally, while the intensive-investment response to technique intensity is concentrated earlier.

The trajectory results support the main timing evidence without carrying the main argument. Expiration-driven peer activity becomes more relevant after 2015, but its response depends on the local signal's information content. Where nearby peer productivity becomes less predictive of cross-sectional productivity differences, peer-performance responses do not weaken. Those formations also continue to become more productive. They do not rise merely because more technique attributes become productivity-related. They persist where the highest-loading technique choices become more predictive. Local networks do not mechanically amplify the national episode. They matter when the local network state is informative for the relevant adoption or investment margin.

OA.4.3 Alternative Explanations and Placebos

A general local drilling boom could increase both horizontal and vertical activity. Table OA.19 replaces the horizontal outcomes with vertical drilling outcomes while retaining the expiration-driven peer-activity measure and network signals. The estimates do not show a parallel positive response in vertical drilling.

A positive interaction need not imply that peer activity or the signal has a positive lower-order coefficient. Table OA.20 reports the peer-activity, signal and interaction coefficients from each pooled specification. The lower-order peer-activity coefficient is the intercept of the marginal response over signal strength and is evaluated outside the support of the positive-signal estimation samples.

Measured network signals may not account for every response to peer activity. Table OA.21 estimates the peer-activity response where none of the four signals is statistically distinguishable from zero and compares this group with the full panel and each signal-specific sample. Peer activity is positively associated with the two extensive outcomes in this small and unusually active comparison group. The intensive-adoption response is also positive at the 10 percent level, while the intensive-investment response is imprecise. The group contains 2.7 percent of the panel and about 6 percent of observed adoption and investment.

Figure OA.4 compares the response across firm size and prior horizontal-production experience. The response gaps change over the episode, although the evidence does not establish whether

changing firm composition explains the aggregate reversal or accompanies it.

Adoption records any upward shift in a firm's horizontal share. A firm can adopt more than once. Table OA.22 separates firms by their horizontal experience before the response quarter and re-estimates the vintage-block responses within each group. The responses differ across prior experience groups, although the estimates do not establish whether changing firm composition explains the aggregate timing pattern.

OA.4.4 Measurement of Peer Activity and Network Signals

The baseline assigns an expiration weight to every well linked to a lease, including later wells drilled after an earlier well may have held the lease. Table OA.23 rebuilds peer activity from the first well on each lease. This restriction removes 58 percent of bunching-weighted peer activity but preserves the principal pooled signs. The period and vintage comparisons are less stable under this restricted measure.

The baseline constructs peer activity from wells linked to leases within 0.25 miles. Table OA.28 varies that radius from 0.10 miles to the unrestricted spatial match. The pooled responses are strongest under tighter links between wells and leases. The table reports extensive adoption and intensive investment. Web Appendix Table W.2 reports all four outcomes.

A peer's drilling and completion attributes become public only after its well is completed. Table OA.24 compares peer activity dated at spud with peer activity dated at completion and describes when responding firms complete their wells relative to these disclosures. Completion dating preserves the pooled signs but reduces precision. Among responding wells, 71 percent are completed at least 60 days after the median peer well is completed.

OA.5 Conceptual Framework Illustrations

Table OA.25 reports the parameter values used in the counterfactual exercise. Figure OA.5 adds the return, attenuation, and starting-cost components sequentially using the same fitted parameters. Figure OA.6 recalibrates the model after removing one component at a time. The peer-performance calibration uses adoption alone because the late-vintage adoption and investment estimates have opposite signs. A common conversion factor would be negative. Table OA.26 reports the five adoption targets.

With adoption alone and two local-development states, the five estimates identify four parameter combinations. Setting $m_1 = 0$ gives $(m_0, b, f, d) = (1.09, +\infty, 4.75, 3.41)$. Setting $m_0 = 0$ gives $(m_1, b, f, d) = (1.09, +\infty, 3.66, 3.14)$. Both normalizations predict -0.14, -0.07, -0.02, 0.23, and -0.01 for the five groups in Table OA.26, with calendar responses of -0.38, 0.26, and 0.02. The five-parameter loss has a flat direction. The same mechanisms match the qualitative peer-performance pattern only at $b = +\infty$, so the saturation parameter has no interior estimate.

Supplemental Figures

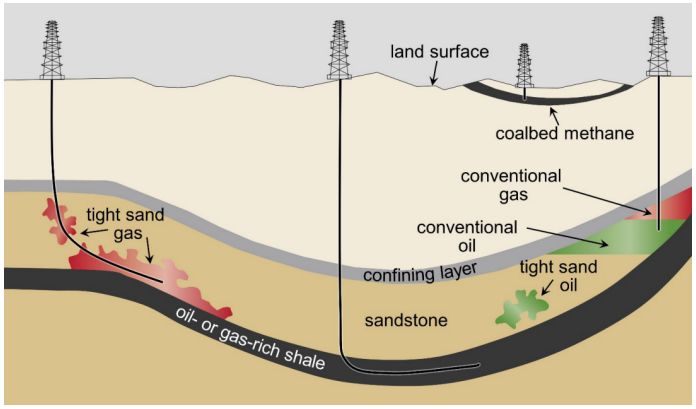


Figure OA.1: Conventional vertical and horizontal extraction

Notes. A conventional vertical well produces from the rock it passes through. A horizontal well bends near the target depth and runs laterally through the shale, fractured at many stages along the lateral, reaching resources a vertical well cannot economically produce. *Source.* U.S. Environmental Protection Agency.

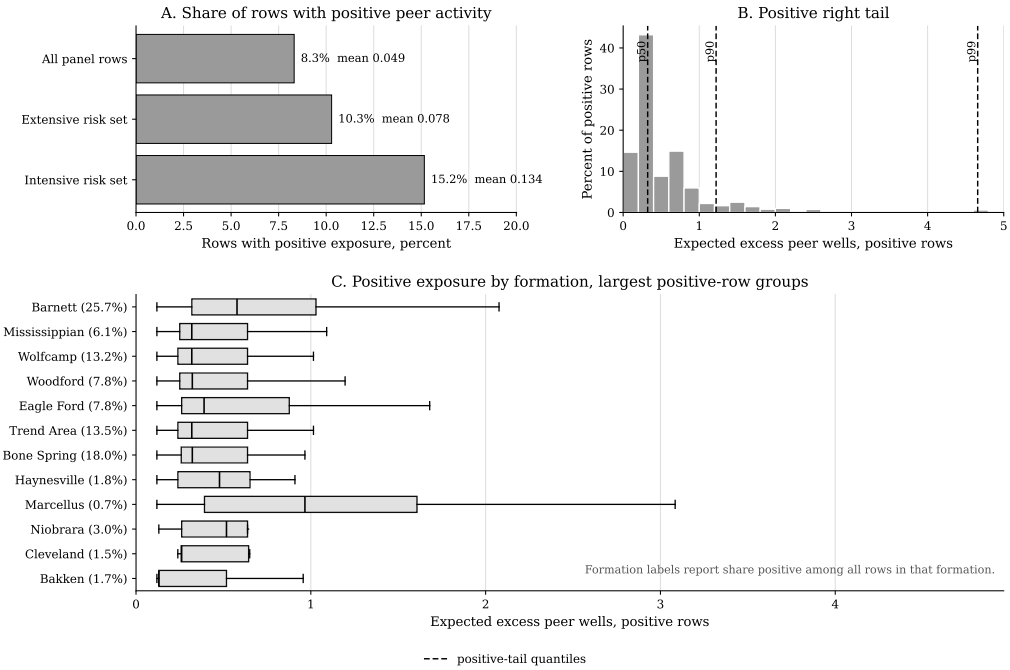


Figure OA.2: Distribution of expiration-driven peer activity

Notes. The figure summarizes the one-quarter-lagged, firm-excluded peer-activity measure in the response panel. The measure sums assigned peer wells in the county-quarter using each well's estimated excess share near lease expiration as its weight. It removes wells operated or leased by the responding firm. Panel A reports the share of observations with positive peer activity and its unconditional mean in the full panel and in the extensive and intensive risk sets. Panel B shows the distribution conditional on positive peer activity. Panel C reports that positive distribution by formation for the formations with the largest positive samples.

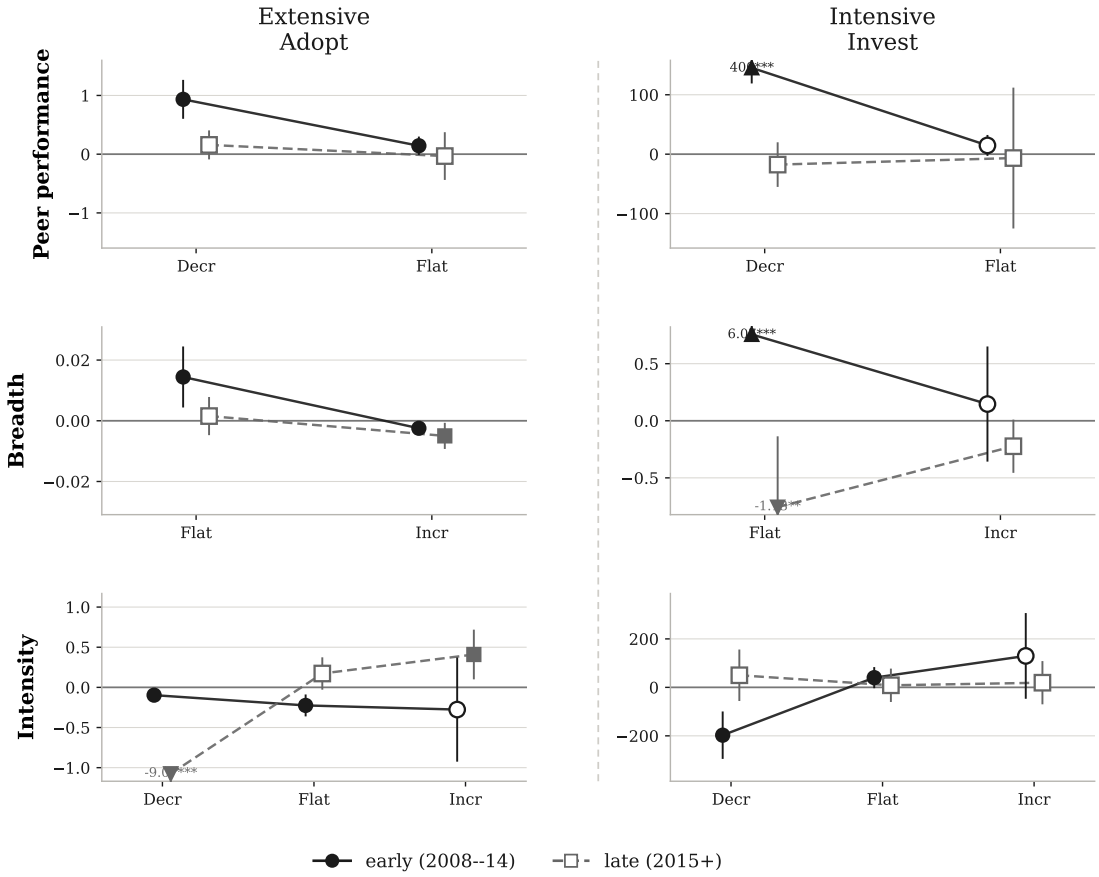


Figure OA.3: Response by the signal's own local trajectory

Notes. The sample applies the same-county own-expiration exclusion. The unit is a firm–county–formation–quarter. Each marker reports the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed network signal. The regressions are estimated separately by the signal's path across successive 2,000-well formation blocks, classified as decreasing, flat, or increasing, and by calendar cohort, 2008–14 or 2015 onward. The outcomes are extensive adoption and horizontal-well counts among firms with positive horizontal investment. Each specification includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Filled markers are statistically different from zero at the 5 percent level. Whiskers are 95 percent confidence intervals based on formation-block-clustered standard errors. Off-scale estimates are printed beside their markers. Unlike Figure 6 in the main text, filled markers here denote significance at the 5 percent level. In the column titles, "Adopt" is extensive adoption and "Invest" is horizontal-well counts among firms with positive horizontal investment. In the row labels, "Breadth" is peer-technique breadth and "Intensity" is technique intensity. On the horizontal axes, "Decr," "Flat" and "Incr" denote decreasing, flat and increasing signal paths.

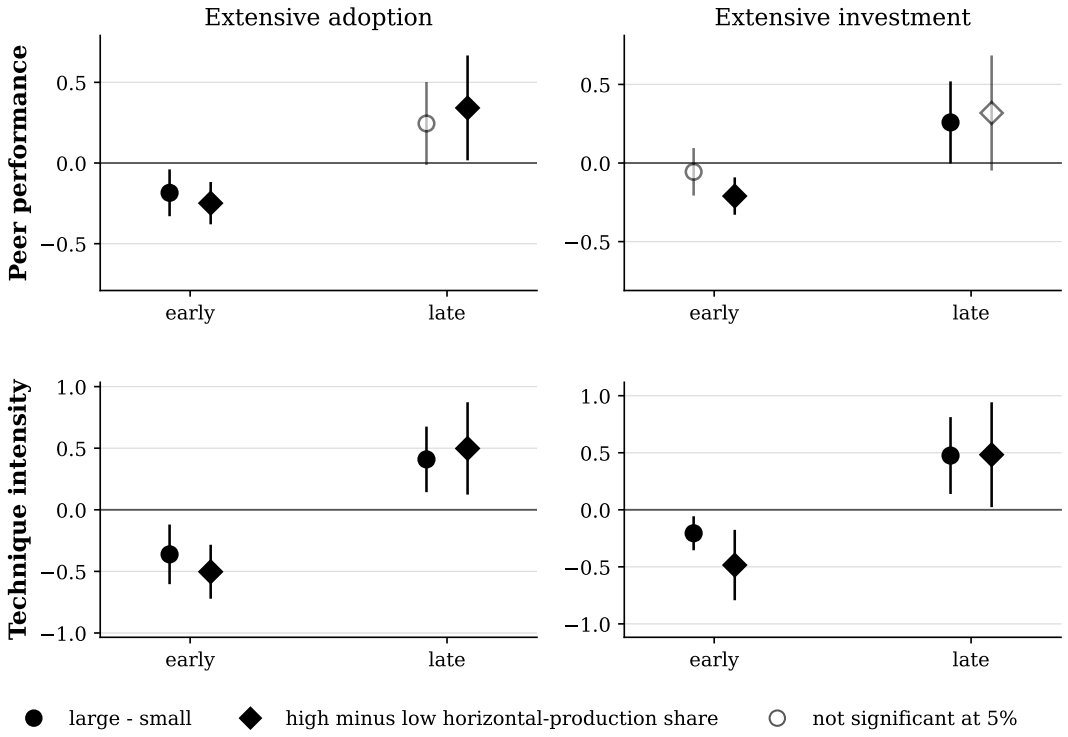


Figure OA.4: Firm-type differences in the response by cohort

Notes. Each point is the high-minus-low difference in the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed network signal. The regressions are estimated separately for 2008–14 and 2015 onward, for peer performance and technique intensity and for the two extensive outcomes. Both signals use distance-time weights. Large firms are classified once using the cross-firm median of firms' average total well stocks over the full sample. Sophisticated firms are classified separately within each cohort using the cross-firm median of firms' average horizontal-production shares. Every regression includes all lower-order interaction terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Each signal-specific sample contains observations for which that signal is positive and statistically different from zero. Positive values indicate a larger response among the high-type firms. Whiskers are 95 percent confidence intervals based on formation-block-clustered standard errors. Circles report large minus small firms. Diamonds report high minus low horizontal-production share. Filled markers are statistically different from zero at the 5 percent level. Hollow markers are not.

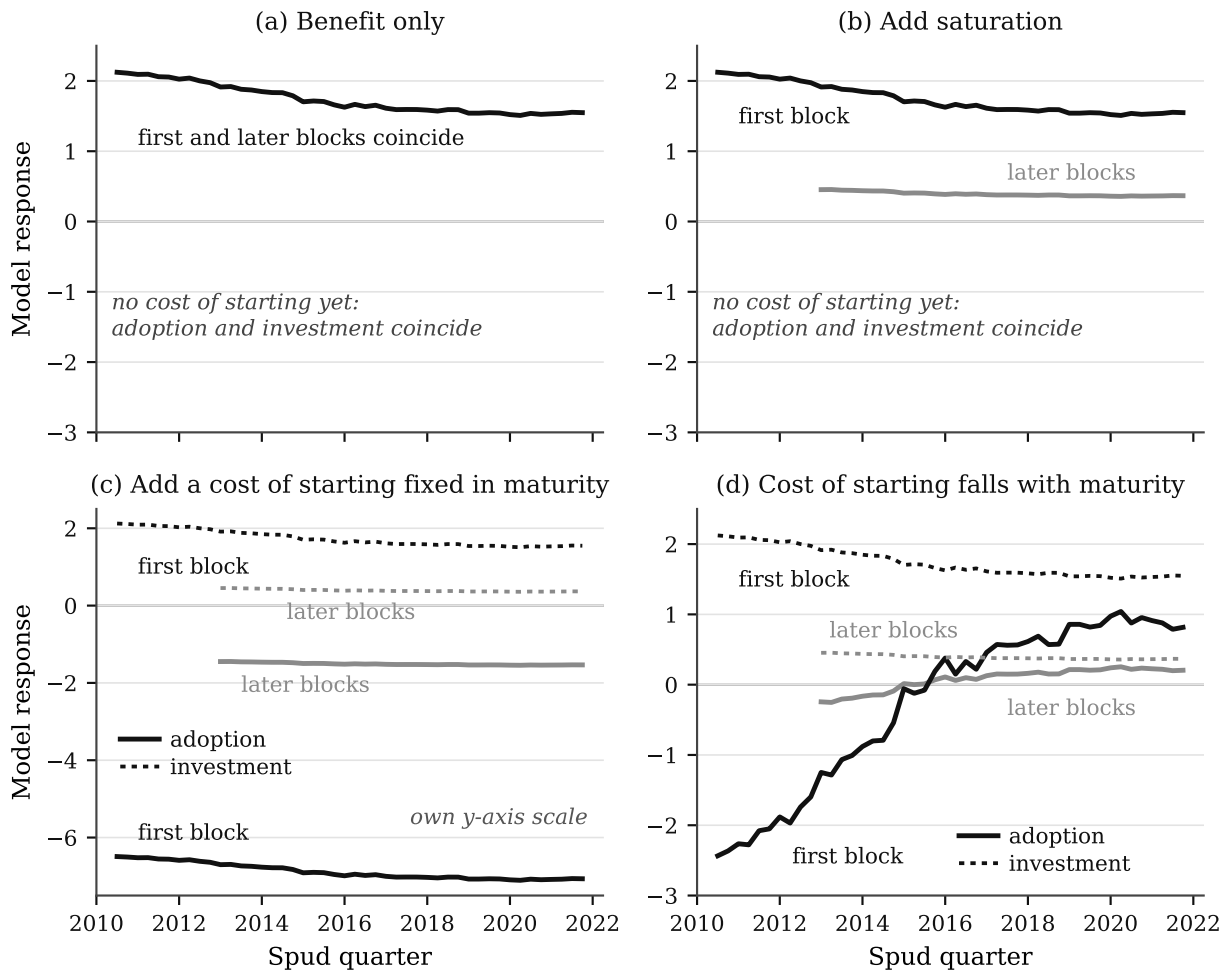


Figure OA.5: Building up the calibrated responses

Notes. Each panel evaluates the model at the industry-wide horizontal share observed each quarter. Black curves hold local development at a formation's first 2,000-well block, $n = 0$. Gray curves hold it at subsequent blocks, $n = 1$. Solid lines report normalized adoption responses. Dotted lines report normalized investment responses. Panel (a) includes only the return response. Panel (b) adds attenuation with local development through b. Panel (c) adds a starting-cost response that varies with local development but not industry maturity. Panel (d) allows that response to vary with both states and reports the full calibrated model. Every panel uses the parameters from the full calibration without recalibration. The horizontal axis, labeled "Spud quarter," is the quarter whose industry-wide horizontal share enters the model. The vertical axis, labeled "Model response," reports the normalized responses. Panel (c) uses a wider vertical scale.

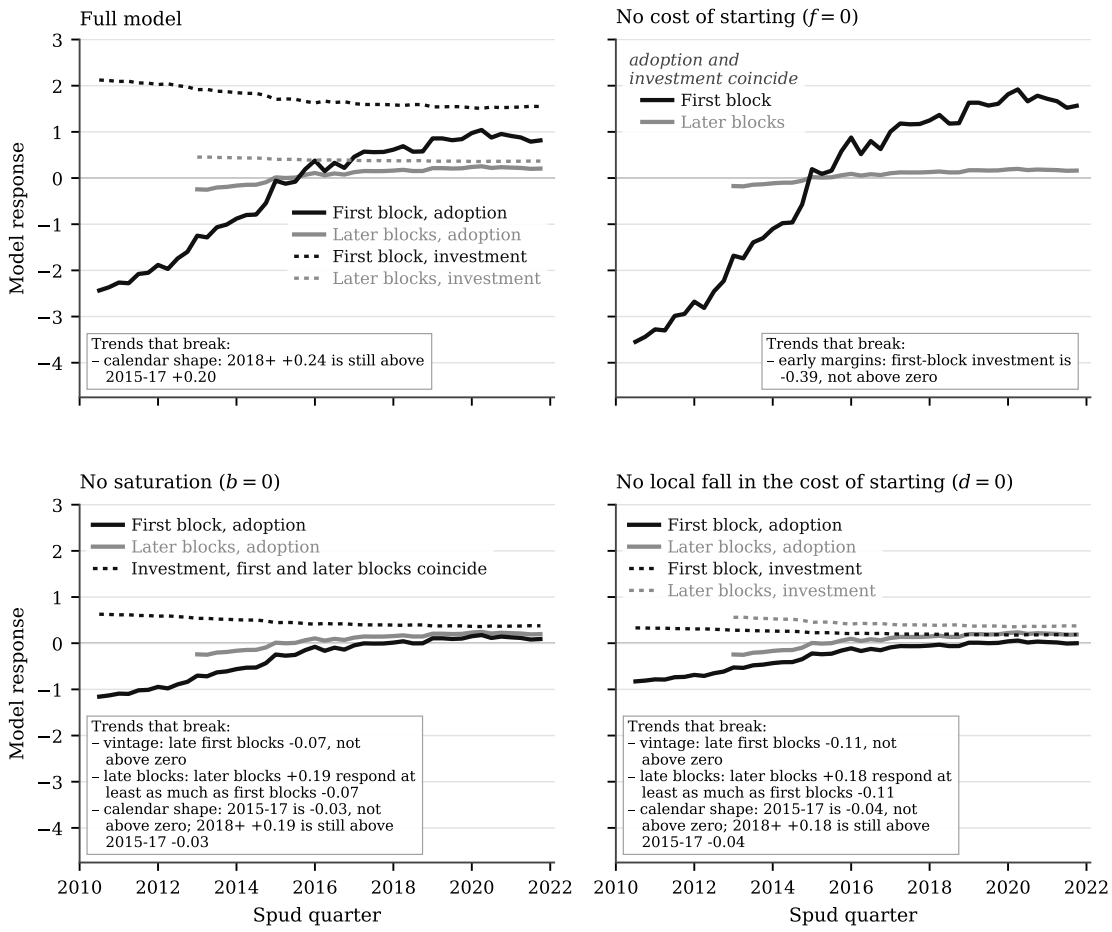


Figure OA.6: Removing one mechanism at a time

Notes. Each panel evaluates a calibrated model at the industry-wide horizontal share observed in each quarter. Black curves hold local development at a formation's first 2,000-well block, $n = 0$. Gray curves hold it at subsequent blocks, $n = 1$. Solid lines report the normalized adoption response and dotted lines report the normalized investment response. Panel (a) is the full model. Panels (b)–(d) set the starting-cost term to zero ($f = 0$), remove local attenuation ($b = 0$), and prevent the starting-cost response from falling with local development ($d = 0$), respectively. Each restricted model is recalibrated to the same ten technique-intensity estimates. Panel text identifies the empirical pattern missed by each restricted model. No empirical estimates are plotted.

Supplemental Tables

Table OA.1: Dispersion around contemporaneous national engineering practice

	Interval-length gap		Spud-to-completion gap		Wells	
	Early	Late	Early	Late	Early	Late
Panel A. Peer-performance sample						
First-block state	0.403	0.355	0.694	0.673	7,932	11,245
Later blocks, through 2014	0.349	–	0.769	–	11,278	–
Later blocks, 2015–16	0.337	–	0.805	–	3,173	–
Later blocks, 2017 onward	0.365	0.313	0.669	0.578	4,725	9,135
Later blocks, 2019 onward	0.356	0.310	0.659	0.581	3,607	5,866
Panel B. Technique-intensity sample						
First-block state	0.268	0.351	0.693	0.670	7,460	7,855
Later blocks, through 2014	0.294	–	0.803	–	9,233	–
Later blocks, 2015–16	0.396	–	0.797	–	4,153	–
Later blocks, 2017 onward	0.365	0.334	0.634	0.606	11,090	16,259
Later blocks, 2019 onward	0.354	0.301	0.642	0.588	5,863	9,877

Notes. The table reports within-group standard deviations of two well-level gaps from contemporaneous national practice. The interval-length gap is the log perforated interval of a horizontal well minus the log median among horizontal wells drilled outside its formation in the same quarter. The spud-to-completion gap is the log number of days from spud to completion minus the value predicted from log interval length among those same-quarter national wells. Each national benchmark requires at least 30 wells. The sample contains wells matched to the same-county own-expiration-restricted response panel with positive interval length and spud-to-completion time between 1 and 3,000 days. The match retains 78,028 of the 169,374 otherwise eligible wells. Completion times for wells spudded near the end of the sample may be selected toward wells completed before the data end. Panel A uses the peer-performance response sample, and Panel B uses the technique-intensity response sample. Early-vintage formations completed their first 2,000-well block by 2012Q3. Late-vintage formations completed it in 2014Q1 or later. The 2019-onward rows are subsets of the 2017-onward rows. Late-vintage formations have no later-block wells before 2017. The last two columns report the number of wells in each cell. The entries are descriptive point estimates without standard errors.

Table OA.2: Expiration bunching conditional on county-quarter drilling activity

Specification	$\tau = -4$	$\tau = -3$	$\tau = -2$	$\tau = -1$	$\tau = 0$	$\tau = 1$	Cells
No FE PPML	0.2705***	0.1221	0.2926**	0.3808***	0.3762***	0.1337	54,111
	(0.0995)	(0.1291)	(0.1458)	(0.1336)	(0.1186)	(0.0851)	
County and quarter FE PPML	0.2705***	0.1221	0.2926**	0.3808***	0.3762***	0.1337	54,111
	(0.0995)	(0.1291)	(0.1458)	(0.1336)	(0.1186)	(0.0851)	
County-quarter FE PPML	0.2705***	0.1221	0.2926**	0.3808***	0.3762***	0.1337	54,111
	(0.0995)	(0.1291)	(0.1458)	(0.1336)	(0.1186)	(0.0851)	
County-quarter FE OLS	0.0528***	0.0240	0.0658**	0.0960***	0.1014***	0.0344*	54,111
	(0.0194)	(0.0227)	(0.0278)	(0.0252)	(0.0246)	(0.0176)	
Full-distance county-quarter FE	0.0783	0.0587	0.0228	0.1286**	0.1759***	0.0276	143,259
	(0.0483)	(0.0500)	(0.0579)	(0.0619)	(0.0504)	(0.0365)	
Wider-window county-quarter FE	0.5012***	0.4453***	0.6616***	0.7345***	0.6522***	0.2860***	67,662
	(0.0822)	(0.1009)	(0.1065)	(0.1012)	(0.0763)	(0.0704)	

Table OA.3: Name-matched lease links and peer-activity responses

Panel A. Well bunching around lease expiration

Event time to expiration	Observed assigned wells	Counterfactual	Excess	Excess / observed (%)
$[-8, -5]$	6,862	6741.1	120.9	1.76
-4	2,267	1835.0	432.0	19.06
-3	2,265	1887.9	377.1	16.65
-2	2,690	1979.4	710.6	26.41
-1	3,394	2133.4	1260.6	37.14
0	6,354	2366.6	3987.4	62.75
1	3,026	2684.3	341.7	11.29
$[2, 5]$	10,513	10503.7	9.3	0.09

Panel B. Pooled response estimates

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance	0.0159*	0.0191	0.4062***	0.4824
	(0.0087)	(0.0116)	(0.1544)	(0.7552)
Peer skill	−0.0088	−0.0120*	−0.0713	2.797**
	(0.0071)	(0.0065)	(0.2076)	(1.405)
Peer-technique breadth	−0.0002**	−0.0002**	0.0002	0.0003
	(0.0001)	(0.0001)	(0.0006)	(0.0021)
Technique intensity	0.0273*	0.0339*	0.2835	1.215
	(0.0159)	(0.0197)	(0.5278)	(1.262)

Notes. In Table OA.2, the unit is a county–quarter–event-time bin, where event time is quarters from lease expiration to spud. The dependent variable is the count of uniquely assigned wells. Expiration-window indicators for $\tau = -4, \ldots, 1$ enter with a fifth-degree event-time polynomial. Rows vary fixed effects, estimator, match distance or estimation window. Standard errors cluster by county, the level at which bins are formed. The Poisson (PPML) coefficients are in log points of the expected well count. The OLS coefficients are in wells per bin. County and county-quarter fixed effects leave the Poisson and OLS estimates unchanged. The estimation panel is a complete rectangle of county-quarter cells by event-time quarters and every regressor is a function of event time alone, so the fixed effects are orthogonal to the regressors. The full-distance and wider-window rows vary the sample. In Table OA.3, Panel A aggregates wells under the operator–lessee name-match construction and compares observed drilling with the fitted profile outside the expiration window. The fitted profile is a fifth-order event-time polynomial estimated over quarters −12 through 4. Panel B reconstructs firm-excluded peer activity from those links and reports its one-quarter-lagged interaction with each signal. The response regressions use the baseline controls, fixed effects, signal-specific samples and same-county own-expiration restriction. Intensive outcomes condition on positive activity. Standard errors cluster by formation-block. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.4: County bootstrap for estimated peer activity

Signal	Outcome	Estimate	Clustered SE	Bootstrap SE	Percentile interval
Peer performance	Extensive adoption	0.1531	0.0649	0.0975	[0.0400, 0.3817]
	Intensive investment	5.6188	6.6970	10.6539	[−12.0137, 28.6767]
Peer skill	Extensive adoption	0.0404	0.1688	0.7767	[−0.4847, 1.0885]
	Intensive investment	−37.4633	15.5702	46.9102	[−106.9768, 58.8724]
Peer-technique breadth	Extensive adoption	−0.0020	0.0006	0.0008	[−0.0039, −0.0008]
	Intensive investment	−0.3267	0.1741	0.2091	[−0.7658, 0.0438]
Technique intensity	Extensive adoption	0.2661	0.0955	0.1457	[0.0265, 0.5800]
	Intensive investment	37.6117	27.1502	19.2656	[4.6418, 80.9920]

Notes. The unit in the response regression is a firm–county–formation–quarter. Each estimate is the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed network signal for extensive adoption or intensive investment. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Signal-specific samples require the listed signal to be positive and statistically different from zero. Intensive investment also requires positive horizontal investment. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. The table reports 499 county-bootstrap draws. Each draw resamples counties, re-estimates the national lease-expiration bunching curve, reconstructs the firm-excluded peer-activity measure and re-estimates the pooled response equation. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. The network signals are held fixed. Clustered SE is the formation-block-clustered standard error reported with the baseline estimate. Bootstrap SE is the standard deviation of the bootstrap estimates. The final column gives the 2.5th and 97.5th percentiles of their distribution.

Table OA.5: Network-signal sensitivity to alternative spatial weights

Signal	Alternative weight	N	Correlation	Sign agreement (%)	Sample agreement (%)
Peer performance	Inverse distance	49	0.869	89.8	85.7
Peer performance	Equidistant	46	0.640	80.4	65.2
Peer performance	Relative skill	49	0.288	67.3	67.3
Peer skill	Inverse distance	49	0.901	79.6	93.9
Peer skill	Equidistant	46	0.767	87.0	89.1
Peer skill	Relative skill	49	0.043	57.1	77.6
Peer-technique breadth	Inverse distance	54	0.517	–	74.1
Peer-technique breadth	Distance-time	54	0.674	–	77.8
Peer-technique breadth	Relative skill	54	0.417	–	63.0
Technique intensity	Inverse distance	37	0.916	–	94.6
Technique intensity	Equidistant	38	0.781	–	97.4
Technique intensity	Relative skill	37	0.500	–	89.2

Notes. Each row compares an alternative spatial weight with the preferred measure using the same completed formation blocks. Distance-time is preferred for peer performance, peer skill and technique intensity. Equidistant is preferred for peer-technique breadth. Relative skill weights peer wells by differences in operator skill and is distinct from the peer-skill signal. N is the number of completed formation blocks with both measures. Correlation is the Pearson correlation between the two signal values. Sign agreement is reported for the signed peer-performance and peer-skill coefficients and omitted for the nonnegative breadth and intensity measures. Sample agreement is the share of common blocks assigned to the same positive-signal sample under both weights. A positive signal requires a positive coefficient significant at the 10 percent level for peer performance and peer skill, at least one positive and statistically distinguishable attribute for breadth, and a joint Wald test significant at the 10 percent level for technique intensity.

Table OA.6: Signal timing buffer

	Extensive adoption		Intensive investment	
	Baseline	Timing buffer	Baseline	Timing buffer
Peer performance	0.1531	0.1850	5.6188	8.7573
	(0.0649)	(0.0540)	(6.6970)	(6.5659)
Peer skill	0.0404	−0.0614	−37.4633	−43.6748
	(0.1688)	(0.2471)	(15.5702)	(20.3489)
Peer-technique breadth	−0.0020	−0.0017	−0.3267	−0.3681
	(0.0006)	(0.0007)	(0.1741)	(0.1793)
Technique intensity	0.2661	0.2377	37.6117	26.7706
	(0.0955)	(0.1008)	(27.1502)	(22.6036)

Table OA.7: Standardized pooled responses to peer activity

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance [distance-time]				
$z(\widehat{E}^{-i}_{c,q-1}) \times z(\text{signal})$	0.0008^{**}	0.0010^{**}	0.1000^{***}	0.0607
	(0.0004)	(0.0004)	(0.0285)	(0.0723)
N	582,531	582,531	7,234	8,565
R^2	0.381	0.402	0.911	0.641
Peer skill [distance-time]				
$z(\widehat{E}^{-i}_{c,q-1}) \times z(\text{signal})$	0.0003	0.0002	−0.0140	-0.3039^{**}
	(0.0013)	(0.0008)	(0.0186)	(0.1263)
N	152,297	152,297	912	1,080
R^2	0.536	0.541	0.980	0.866
Peer-technique breadth [equidistant]				
$z(\widehat{E}^{-i}_{c,q-1}) \times z(\text{signal})$	-0.0021^{***}	-0.0027^{***}	−0.0396	-0.5919^{*}
	(0.0007)	(0.0008)	(0.0654)	(0.3155)
N	683,440	683,440	7,058	8,420
R^2	0.354	0.370	0.957	0.579
Technique intensity [distance-time]				
$z(\widehat{E}^{-i}_{c,q-1}) \times z(\text{signal})$	0.0009^{***}	0.0010^{**}	0.0417	0.1845
	(0.0003)	(0.0004)	(0.0345)	(0.1332)
N	778,002	778,002	8,704	10,348
R^2	0.358	0.378	0.862	0.615
Controls	Yes	Yes	Yes	Yes
Operator / county / operator-county / quarter FE	Yes	Yes	Yes	Yes

Notes. The unit in both tables is a firm–county–formation–quarter. Each entry is the coefficient on one-quarter-lagged, firm-excluded peer activity interacted with the listed signal. Table OA.6's timing-buffer columns drop the first response quarter after each 2,000-well block closes, removing 119,687 of 1,002,519 panel rows before signal-specific restrictions. The extensive-adoption timing-buffer samples contain 513,727, 139,638, 606,573 and 681,275 observations for peer performance, peer skill, peer-technique breadth and technique intensity. The corresponding intensive-investment samples contain 6,978, 936, 6,915 and 8,352 observations. Baseline sample sizes are those of main Table 2. Signals are not re-estimated. Table OA.7 instead replaces peer activity and the signal with within-sample z-scores, so its coefficient is the interaction of one-standard-deviation changes. Rescaling leaves R^2 unchanged. Both tables use the baseline lower-order terms, controls, fixed effects, signal-specific samples and same-county own-expiration restriction. Intensive investment conditions on positive horizontal investment. Standard errors cluster by formation-block. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.8: Response to peer activity by attributes reported

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Panel A. All rows				
Peer activity × attributes reported	−0.0011	−0.0013	0.0351	−0.1261
	(0.0012)	(0.0014)	(0.0357)	(0.1123)
Observations	1,002,243	1,002,243	11,808	14,078
Panel B. Technique-intensity sample				
Peer activity × attributes reported	0.0031**	0.0039**	0.1347	0.5388*
	(0.0015)	(0.0017)	(0.1874)	(0.2866)
Observations	778,002	778,002	8,704	10,348

Notes. The unit is a firm–county–formation–quarter. The coefficient is on one-quarter-lagged expiration-driven peer activity interacted with the average number of drilling and completion attributes reported per horizontal well in the formation-block whose network signal the observation uses. Each coefficient is the change in the outcome per estimated excess peer well per additional attribute reported. Every regression includes the lower-order peer-activity and attribute-reporting terms. The specification otherwise follows the pooled response specification, with the full control set, operator, county, operator-by-county and quarter fixed effects and the same-county own-expiration restriction. Panel A uses every observation. Panel B uses the technique-intensity sample, where that signal is positive and statistically different from zero. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive adoption is the size of that change conditional on positive adoption. Extensive investment indicates positive horizontal drilling. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Panel A drops 276 singleton observations absorbed by the fixed effects. Every panel row has an attribute-reporting measure.

Table OA.9: Peer-technique breadth and attribute redundancy

Breadth path	Formations	Blocks	Breadth	New attrs.	Redundancy	Abs. corr.
Increasing	8	26	2.50	2.36	1.17	0.64
Flat	4	22	1.45	0.95	0.79	0.64
Unclassified	3	6	4.50	4.50	1.62	0.54

Notes. The unit is a formation by completed 2,000-well development block. The table is descriptive. It compares blocks in formations where peer-technique breadth increases across successive blocks with blocks in formations where breadth remains flat. Breadth is the number of observable peer-technique attributes whose productivity coefficient is positive and statistically different from zero at the 5 percent level under the preferred equidistant weight. New attributes satisfy this criterion in the current block but not in an earlier block from the same formation. Redundancy divides the sum of the attributes' separate explanatory contributions by their joint explanatory contribution. Larger values indicate that the attributes overlap more in what they explain. Abs. corr. is their mean absolute pairwise correlation within the block. Breadth paths are classified by visual inspection of each formation's block-by-block breadth. Unclassified formations are Cleveland, which has a single block, Fayetteville, whose breadth is zero in both blocks, and Mississippian, whose breadth falls from 11 to 0 and then 1.

Table OA.10: Peer activity and own well design

	L1	*L2*	*L3*	*L4*
Panel A. Own perforated interval (ln ft)				
Peer performance (N=32,223)	-0.7921^{*}	-0.1919	0.0898	-0.7179^{***}
	(0.3851)	(0.3916)	(0.1876)	(0.1602)
Peer skill (N=6,041)	1.8220^{**}	1.7607^{*}	2.7432^{**}	2.4293^{**}
	(0.3818)	(0.5629)	(0.5884)	(0.4767)
Peer-technique breadth (N=35,619)	0.0040	0.0218^{***}	0.0120^{*}	0.0023
	(0.0103)	(0.0068)	(0.0059)	(0.0052)
Technique intensity (N=44,460)	-0.0937	1.0234	-0.0950	0.4523
	(0.8768)	(0.6701)	(0.5365)	(0.6441)
Panel B. Own drill time (days)				
Peer performance (N=32,703)	-430.3124^{**}	146.2428	232.1192	-64.1576
	(206.3184)	(94.7101)	(136.0729)	(111.0667)
Peer skill (N=5,750)	-267.3334	1096.9645^{**}	-1244.1648^{**}	-803.9316^{***}
	(162.6886)	(340.5243)	(216.9755)	(91.2761)
Peer-technique breadth (N=35,950)	2.6639	4.4747	2.1810	0.6792
	(2.7347)	(4.0829)	(3.6359)	(2.0898)
Technique intensity (N=44,770)	1303.0812	619.7703	784.5314^{**}	271.7350
	(836.4630)	(488.5318)	(361.3846)	(338.9526)
Firm-quarter controls	Yes	Yes	Yes	Yes
County-quarter controls	Yes	Yes	Yes	Yes
Operator / county / operator-county / quarter FE	Yes	Yes	Yes	Yes
Standard errors	formation-block	formation-block	formation-block	formation-block

Notes. The unit is a horizontal well drilled from 2014Q4 through 2022Q4 and matched to the firm–county–formation–quarter response panel. Each row reports an interaction coefficient from a regression that jointly includes lags one through four of expiration-driven peer activity interacted with the listed network signal. L1 through L4 denote peer activity lagged one through four quarters. The corresponding lower-order peer-activity and signal terms are included. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Each signal-specific sample contains wells for which that signal is positive and statistically different from zero under its preferred weight. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Perforated interval is the log length in feet of the completed horizontal section that is fractured. Drill time is the number of days from spud to completion. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. Standard errors are clustered by formation-block and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.11: Headline pooled responses with lags one through four

	Extensive adoption	Intensive investment
Peer performance [distance-time]		
$L1$	0.1103^{**}	10.8842^{**}
	(0.0439)	(4.5347)
$L2$	0.0332	6.4468
	(0.0755)	(8.6082)
$L3$	0.1392^{***}	-17.6677
	(0.0392)	(14.3150)
$L4$	0.0905	-32.1251
	(0.0675)	(27.2803)
Technique intensity [distance-time]		
$L1$	0.2477^{***}	40.8280
	(0.0884)	(28.6120)
$L2$	0.0257	3.7816
	(0.0712)	(11.7856)
$L3$	0.0663	1.4570
	(0.0883)	(15.3864)
$L4$	-0.0107	-28.9124
	(0.0900)	(24.3881)

Notes. The unit is a firm–county–formation–quarter. Rows report the coefficient on $\widehat{E}^{-i}_{c,q-k} \times S_{gq}$, with lags $k = 1, \ldots, 4$ entered jointly. S_{gq} is the listed network signal. L1 through L4 denote peer activity lagged one through four quarters. Each regression includes the corresponding lower-order peer-activity terms, the signal term, baseline controls and fixed effects, the signal-specific sample and the same-county own-expiration restriction. Intensive investment conditions on positive horizontal investment. Both signals use distance-time weights. Standard errors cluster by formation-block. Web Appendix Table W.6 reports all four signals and all four outcomes. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.12: Period responses under alternative control specifications

Signal	Period / contrast	Era-specific controls		Common controls	
		Extensive adoption (1)	Intensive investment (2)	Extensive adoption (3)	Intensive investment (4)
Peer performance	2008–14	−0.046	10.317	0.094	15.100*
		(0.057)	(9.463)	(0.078)	(8.626)
	2015–17	0.282***	11.013	0.286***	2.247
		(0.078)	(12.761)	(0.067)	(24.274)
	2018+	−0.327*	66.071***	−0.063	37.664*
		(0.194)	(20.803)	(0.191)	(21.805)
	Rise: (2015–17) − (2008–14)	0.328***	0.696	0.192*	−12.853
		(0.097)	(16.029)	(0.105)	(27.111)
	Fall: (2018+) − (2015–17)	−0.610***	55.058**	−0.349	35.417
		(0.194)	(24.988)	(0.209)	(34.508)
Technique intensity	2008–14	−0.174***	29.324	0.228	42.842**
		(0.035)	(22.499)	(0.174)	(19.206)
	2015–17	0.357*	−18.525	0.281*	−5.049
		(0.211)	(25.286)	(0.150)	(28.581)
	2018+	0.280*	−24.868	0.322**	−27.489
		(0.145)	(52.654)	(0.144)	(51.627)
	Rise: (2015–17) − (2008–14)	0.531**	−47.849	0.053	−47.890
		(0.215)	(32.579)	(0.184)	(37.199)
	Fall: (2018+) − (2015–17)	−0.077	−6.343	0.041	−22.441
		(0.245)	(57.570)	(0.201)	(62.392)
Standard errors		formation-block			

Notes. The unit is a firm–county–formation–quarter. Columns 1 and 2 report the preferred specification, which allows the control coefficients and operator, county, operator-by-county and calendar-quarter fixed effects to vary across eras. Columns 3 and 4 hold those coefficients and fixed effects common across eras while allowing the peer-activity, signal and interaction coefficients to vary. Each entry is the coefficient on one-quarter-lagged, firm-excluded expiration-driven peer activity interacted with the listed network signal. All regressions include the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. Intensive investment conditions on positive horizontal investment. Both signals use distance-time weights. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. The rise and fall rows are directly tested contrasts. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.13: Additional responses by period

Signal	Period / contrast	Unplotted breadth outcomes: Extensive adoption	Unplotted breadth outcomes: Intensive investment	Additional margins: Extensive investment	Additional margins: Intensive adoption
Peer performance	2008–14	—	10.317	0.0009	1.715
			(9.463)	(0.0702)	(1.776)
	2015–17	—	11.013	0.2807***	15.422
			(12.761)	(0.0919)	(9.726)
	2018+	—	66.071***	−0.3430	−0.236
			(20.803)	(0.2608)	(3.415)
	Rise: (2015–17) − (2008–14)	—	0.696	0.2798**	13.707
			(16.029)	(0.1167)	(9.241)
	Fall: (2018+) − (2015–17)	—	55.058**	−0.6237**	−15.658
			(24.988)	(0.2521)	(10.396)
Technique intensity	2008–14	—	29.324	−0.1530**	4.782
			(22.499)	(0.0740)	(3.779)
	2015–17	—	−18.525	0.2962	7.995
			(25.286)	(0.2427)	(15.994)
	2018+	—	−24.868	0.3191**	4.019
			(52.654)	(0.1599)	(27.112)
	Rise: (2015–17) − (2008–14)	—	−47.849	0.4491*	3.213
			(32.579)	(0.2512)	(16.693)
	Fall: (2018+) − (2015–17)	—	−6.343	0.0229	−3.976
			(57.570)	(0.2773)	(31.621)
Peer-technique breadth	2008–14	−0.0006	0.213	−0.0009*	−0.035
		(0.0005)	(0.159)	(0.0005)	(0.027)
	2015–17	−0.0007	−0.397	−0.0007	−0.238
		(0.0006)	(0.405)	(0.0007)	(0.159)
	2018+	−0.0030*	−0.283	−0.0039**	0.031
		(0.0018)	(0.187)	(0.0019)	(0.033)
	Rise: (2015–17) − (2008–14)	−0.0000	−0.610	0.0002	−0.203
		(0.0008)	(0.432)	(0.0010)	(0.162)
	Fall: (2018+) − (2015–17)	−0.0024	0.114	−0.0031	0.269
		(0.0019)	(0.446)	(0.0020)	(0.165)
Firm-quarter and county-quarter controls			Yes		
Era-specific fixed effects and controls			Yes		
Standard errors			formation-block		

Notes. The first outcome column reports peer-technique breadth's extensive-adoption response, which is not plotted in the main-paper period figure. Dashes suppress the corresponding peer-performance and technique-intensity estimates. The intensive-investment column reports all three signals, including the peer-performance and technique-intensity estimates shown in that figure. The final two columns report extensive investment and intensive adoption for all three signals. The interaction coefficient β is estimated at the one-quarter lag, separately by period, under the separate-fit specification of the main paper, in which the control coefficients are estimated within each period. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects within each period. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. The rise and fall rows are directly tested contrasts. Standard errors clustered by formation-block are in parentheses. Each specification is estimated on the formation-block observations in which its own signal is positive and individually significant, which is why sample sizes differ across signals. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.14: Own-expiration exclusions in pooled and vintage-block estimates

Panel A. Pooled response, excluding any own expiration

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance	0.1423**	0.1754**	9.965***	−3.655
	(0.0632)	(0.0743)	(2.559)	(8.671)
Peer skill	0.0459	0.0288	−2.493	−41.14***
	(0.1167)	(0.0444)	(2.398)	(10.55)
Peer-technique breadth	−0.0019***	−0.0025***	−0.0363	−0.3221*
	(0.0006)	(0.0007)	(0.0440)	(0.1770)
Technique intensity	0.2392***	0.2835**	13.86	34.79
	(0.0799)	(0.1112)	(10.05)	(34.27)

Panel B. Pooled response, exclusion over the full bunching window

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance	0.142**	0.179**	10.66***	6.389
	(0.0598)	(0.0721)	(2.813)	(6.944)
Peer skill	0.0799	0.0621	−2.588	−43.06***
	(0.264)	(0.193)	(2.076)	(15.51)
Peer-technique breadth	−0.0020***	−0.0024***	−0.0193	−0.248
	(0.0006)	(0.0007)	(0.0361)	(0.177)
Technique intensity	0.247***	0.286**	11.51	34.41
	(0.0901)	(0.122)	(8.902)	(25.38)

Panel C. Vintage-block estimates, exclusion over the full bunching window

	Technique intensity		Peer performance	
	Extensive adoption	Intensive investment	Extensive adoption	Intensive investment
Early vintage, first block	−0.493**	64.34	−0.163**	−13.93**
	(0.229)	(85.80)	(0.0794)	(5.533)
Early vintage, later blocks, ≤2014	−0.176**	33.57***	−0.0712	36.29**
	(0.0743)	(12.10)	(0.0632)	(15.99)
Early vintage, later blocks, 2015+	0.219*	−43.34**	0.197	−70.02
	(0.115)	(18.46)	(0.291)	(70.72)
Late vintage, first block	0.783	29.81	0.206**	40.42**
	(0.565)	(149.23)	(0.0870)	(19.98)
Late vintage, later blocks, 2017+	0.248*	−22.39	−0.690**	113.94*
	(0.142)	(80.65)	(0.286)	(63.56)

Panel D. First-block vintage contrast, late minus early, extensive adoption

	Same county, response quarter (baseline)	Any county, response quarter	Same county, full window
Technique intensity	1.422*	1.456*	1.401*
	(0.774)	(0.772)	(0.732)
Peer performance	0.320**	0.351***	0.309***
	(0.124)	(0.108)	(0.119)
Share of panel rows dropped	0.31%		1.03%
Share of positive-peer-activity rows dropped	0.67%		2.18%

Notes. The unit is a firm–county–formation–quarter. Each response coefficient is on one-quarter-lagged, firm-excluded expiration-driven peer activity interacted with the listed network signal. Panel A follows the pooled specification after dropping every observation from a firm-quarter in which the firm has any matched lease expiration, regardless of county. Panels B and C instead drop firm-county-quarters with a matched same-county lease expiring from one quarter before through four quarters after the response quarter. Panel B is pooled and Panel C follows the vintage-block specification. The Panel C early-vintage later-block 2015+ row combines the finer calendar groups of main Table 3 and is estimated as one group. Panel D reports the first-block late-minus-early vintage contrast. All regressions include the lower-order peer-activity and signal terms, the main controls and operator, county, operator-by-county and quarter fixed effects, and use the corresponding signal-specific samples. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The baseline response-quarter same-county, portfolio-wide and full-window same-county restrictions remove 0.31, 13.3 and 1.03 percent of the full panel, respectively. Web Appendix Table W.4 reports the sample accounting for the same-county and portfolio-wide (any-firm-quarter) restrictions. Panel D estimates the first-block late-minus-early contrast from a regression on first-block observations only, with one set of fixed effects and controls shared by both vintages. Table 3 of the main paper instead reports the contrast from the saturated specification with cell-specific fixed effects, so its baseline values (1.286 and 0.378) differ from the baseline column here.

Table OA.15: Lease exercise and local conditions

Panel A. New leases signed in the county-quarter

	(1)	(2)	(3)	(4)
Expiration-driven peer activity, quarter q	143.8	145.6	67.3	68.1
	(96.6)	(98.0)	(45.9)	(46.6)
Expiration-driven peer activity, quarter $q+1$			75.1*	78.0*
			(40.0)	(42.1)
Expiration-driven peer activity, quarter $q-1$			68.7	67.8
			(50.2)	(49.5)
Scheduled lease expirations, quarter q		0.0835***		0.0845***
		(0.0236)		(0.0269)
County-quarters	17,181	17,181	16,683	16,683
Counties	249	249	249	249
Mean of outcome	73.3	73.3	74.3	74.3
Fixed effects		County and quarter		

Panel B. Log first-year output of wells drilled in the expiration window

	(1)	(2)	(3)	(4)	(5)	(6)
Exercise rate	−0.0249**	−0.0096	−0.0292***	−0.0129		
	(0.0112)	(0.0075)	(0.0099)	(0.0078)		
Share of at-risk leases drilled in window					−0.1342	−0.2219
					(0.3103)	(0.4574)
Wells	3,545	3,286	3,545	3,286	3,545	3,286
Counties	165	139	165	139	165	139
Formation and quarter fixed effects	Yes	No	Yes	No	Yes	No
County-formation and quarter fixed effects	No	Yes	No	Yes	No	Yes
Well-type fixed effects	No	No	Yes	Yes	No	No

Notes. Panel A uses a county-quarter panel of 249 counties from 2005Q1 through 2022Q1. The outcome is the number of new leases signed in the county-quarter. Expiration-driven peer activity is the county-quarter sum of bunching weights over wells linked to a lease within 0.25 miles, before the responding firm's own wells are removed. Scheduled lease expirations count the leases whose primary term ends in the quarter. Panel B uses wells linked to a lease within 0.25 miles and spudded between four quarters before and one quarter after that lease's expiration. The outcome is the log of the well's first twelve months of output in barrels of oil equivalent. The exercise rate is the number of such wells in a county and expiration quarter divided by the number of leases expiring there with no linked well spudded before the window. Columns 5 and 6 replace it with the share of those leases drilled in the window. Well-type fixed effects distinguish horizontal from vertical wells. Panel A coefficients are new leases per estimated excess peer well. Standard errors clustered by county, the level at which expirations are scheduled, are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.15 (continued): Lease exercise and local conditions

Panel C. Expiration-driven drilling and recent productivity

	Excess expiration-driven wells (1)	Any expiration-driven drilling (2)	Wells per scheduled expiration (3)
Recent productivity innovation	-0.0071^{***}	-0.0089^{***}	-0.0007^{*}
	(0.0024)	(0.0033)	(0.0003)
Mean of outcome	0.0941	0.1436	0.0056
County fixed effects	Yes	Yes	Yes
Quarter-by-expiration-band fixed effects	Yes	Yes	Yes
Scheduled-expiration control	Yes	Yes	Yes
Observations	10,955	10,955	10,955
Counties	198	198	198

Notes. The sample contains county-quarters with at least one scheduled lease expiration and observed county productivity in the preceding two quarters. For outcome quarter q, recent productivity innovation is the change from quarter $q-2$ to quarter $q-1$ in the county-quarter effect estimated from decline-adjusted well output after controlling for firm effects. The variable is standardized within the estimation sample. Column 1 uses the county-quarter sum of the bunching weights assigned to wells linked to expiring leases. Column 2 is an indicator for positive expiration-driven drilling. Column 3 divides excess expiration-driven wells by the number of leases scheduled to expire. All specifications control for the log number of scheduled expirations and include county fixed effects. Quarter-by-expiration-band fixed effects compare counties in the same quarter with similar numbers of scheduled expirations. The bands are 1, 2–4, 5–9, 10–24, 25–49, 50–99 and 100 or more leases. Standard errors clustered by county, the level at which expirations are scheduled, are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.16: Oil-price controls and price interactions

Signal	Term	Extensive adoption
Peer performance	Peer activity × signal, with price controls	0.1596^{**}
		(0.0693)
Peer performance	Peer activity × signal × WTI	-0.0275
		(0.0538)
Peer performance	Peer activity × signal × vintage	-0.0486
		(0.0517)
Technique intensity	Peer activity × signal, with price controls	0.2635^{***}
		(0.0963)
Technique intensity	Peer activity × signal × WTI	0.0050
		(0.0849)
Technique intensity	Peer activity × signal × vintage	0.2056
		(0.2137)
Peer-technique breadth	Peer activity × signal, with price controls	-0.0018^{***}
		(0.0006)
Peer-technique breadth	Peer activity × signal × WTI	-0.0001
		(0.0003)
Peer-technique breadth	Peer activity × signal × vintage	0.0002
		(0.0003)

Notes. The unit is a firm–county–formation–quarter and the outcome is extensive adoption. Rows labeled peer activity × signal, with price controls, report the coefficient on $\widehat{E}^{-i}_{c,q-1} \times S_{gq}$ after adding peer activity interacted with the standardized quarterly West Texas Intermediate (WTI) crude oil price. Rows labeled peer activity × signal × WTI report the triple interaction from a regression containing all lower-order terms. The vintage rows replace WTI with standardized formation vintage. Formation vintage is the quarter in which the formation completed its first 2,000-well block, standardized over panel rows. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Each signal-specific sample requires the listed signal to be positive and statistically different from zero. Peer performance and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.17: Network signals and pre-2014 formation gas share

	2008–14	2015–17	2018 onward
Peer performance	−0.771	−0.647	0.087
	(9)	(12)	(15)
Technique intensity	−0.444	−0.297	0.035
	(7)	(11)	(14)

Notes. Each entry is the cross-formation correlation between the listed network signal and a fixed formation gas share for the indicated period. The gas share is calculated from horizontal-well production through 2013. Gas is converted to barrels of oil equivalent at six thousand cubic feet per barrel. Signals are averaged within formation and period before calculating the correlation. Parentheses report the number of formations with nonmissing values for both measures. A negative correlation indicates that stronger signals are associated with formations whose pre-2014 production was more oil intensive.

Table OA.18: Signal levels and well productivity in first blocks by formation vintage

	Early-vintage first blocks	Late-vintage first blocks
Technique intensity	0.0267	0.0197
	(0.0154)	(0.0110)
Peer performance	0.0501	0.0442
	(0.0190)	(0.0264)
Well productivity	0.568	0.186
	(0.932)	(1.102)
Log first-year output	11.373	11.253
	(0.932)	(1.091)

Notes. The unit is a firm–county–formation–quarter in a formation's first completed 2,000-well development block. The table reports the mean signal value over response-panel observations, with standard deviations in parentheses. Early-vintage formations complete their first block by 2012Q3. Late-vintage formations complete it in 2014Q1 or later. Each signal is summarized over observations for which that signal is positive and statistically different from zero under its preferred distance-time weight. The means weight formation blocks by their numbers of response-panel observations instead of weighting formations equally. The two productivity rows instead average over the 14,000 horizontal wells that make up each vintage's first blocks. Well productivity is the well's average log ratio of monthly output to a common hyperbolic decline curve, the outcome of the network estimates. It is measured in log points relative to that curve. First-year output is the log of the well's first twelve months of output in barrels of oil equivalent.

Table OA.19: Vertical-drilling placebo

	Vertical county (ext)	Vertical county count
Peer performance	−0.1355**	−1.324***
	(0.0633)	(0.3356)
Peer skill	0.1055	0.2876
	(0.0803)	(0.7054)
Peer-technique breadth	0.0007	0.0045
	(0.0006)	(0.0036)
Technique intensity	−0.1199	−1.139
	(0.1399)	(0.7027)

Notes. The unit is a firm–county–formation–quarter. The dependent variables are vertical drilling outcomes measured at the firm–county–quarter level. Vertical county extensive is an indicator for positive vertical investment in the firm–county–quarter, and vertical county count is the corresponding vertical well count. The regressor of interest is the interaction between the expiration-driven peer-activity measure and the listed local network signal, measured before the response quarter. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Signal-specific samples require the listed signal to be positive and statistically different from zero. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Standard errors are clustered by formation-block and shown in parentheses. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.20: Lower-order terms in the pooled response specification

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance [distance-time]				
Peer activity	-0.0068^{*}	-0.0074	-0.5016^{**}	-0.1997
	(0.0038)	(0.0044)	(0.1913)	(0.4412)
Signal	0.0921	0.1016	-4.6797^{**}	-30.4128^{**}
	(0.1107)	(0.1238)	(1.9357)	(13.9067)
Peer activity × signal	0.1531^{**}	0.1835^{**}	9.4487^{***}	5.6188
	(0.0649)	(0.0756)	(2.6951)	(6.6970)
N	582,531	582,531	7,234	8,565
Peer skill [distance-time]				
Peer activity	-0.0079	-0.0051	-0.1183	-1.8250
	(0.0345)	(0.0208)	(0.2869)	(3.6747)
Signal	0.0074	0.1256^{**}	–	–
	(0.0277)	(0.0318)		
Peer activity × signal	0.0404	0.0272	-1.7864	-37.4633^{**}
	(0.1688)	(0.1014)	(2.3706)	(15.5702)
N	152,297	152,297	912	1,080
Peer-technique breadth [equidistant]				
Peer activity	0.0131^{***}	0.0170^{***}	0.0002	1.8470^{***}
	(0.0042)	(0.0049)	(0.0561)	(0.4328)
Signal	-0.0009^{***}	-0.0011^{***}	0.0006	-0.0611
	(0.0003)	(0.0004)	(0.0056)	(0.0618)
Peer activity × signal	-0.0020^{***}	-0.0025^{***}	-0.0222	-0.3267^{*}
	(0.0006)	(0.0007)	(0.0367)	(0.1741)
N	683,440	683,440	7,058	8,420
Technique intensity [distance-time]				
Peer activity	-0.0076^{***}	-0.0076^{*}	-0.0883	-0.9504
	(0.0028)	(0.0040)	(0.2869)	(0.9070)
Signal	0.0697	0.0873	4.2210	7.5661
	(0.0707)	(0.0885)	(6.5340)	(9.4924)
Peer activity × signal	0.2661^{***}	0.2988^{**}	8.8524	37.6117
	(0.0955)	(0.1249)	(7.3287)	(27.1502)
N	778,002	778,002	8,704	10,348

Notes. The unit is a firm–county–formation–quarter. The table reports all three coefficients from the pooled one-quarter response specifications underlying the main paper's pooled response table. The peer-activity and signal rows are the lower-order terms in the interacted specification. Their coefficients are evaluated at a signal or peer-activity value of zero. The interaction row is the coefficient reported in the main table. Extensive adoption indicates a shift in the firm's technology mix toward horizontal wells. Extensive investment indicates positive horizontal drilling. Intensive adoption is the change in the firm's horizontal share conditional on positive adoption. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Signal-specific samples require the listed signal to be positive and statistically different from zero. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Standard errors are clustered by formation-block and shown in parentheses. Dashes indicate coefficients absorbed in the intensive peer-skill samples. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.21: Peer activity where no network signal is measurable

Panel A. Response estimates

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
No measurable signal	0.0113***	0.0108**	0.1995*	1.7065
	(0.0042)	(0.0045)	(0.1035)	(1.6684)
N	26,567	26,567	655	798

Panel B. Sample composition

Sample	N	County–formation years N	Adoption rate (percent)	Total adoptions N	Investment rate (percent)	Total investments N	Mean peer activity
Full panel	1,002,519	3,221	1.21	12,116	1.44	14,410	0.049
No measurable signal	26,852	178	2.78	746	3.37	905	0.157
Peer performance	583,019	2,092	1.29	7,503	1.52	8,861	0.051
Peer skill	152,673	471	0.66	1,008	0.78	1,185	0.033
Peer-technique breadth	683,705	2,257	1.07	7,338	1.28	8,739	0.044
Technique intensity	778,269	2,574	1.15	8,950	1.37	10,625	0.036

Notes. The unit in both panels is a firm–county–formation–quarter. In Panel A, the sample is restricted to observations in which none of the preferred peer-performance, peer-skill, peer-technique breadth or technique-intensity signals is statistically distinguishable from zero in either direction. The row reports the coefficient on $\widehat{E}^{-i}_{c,q-1}$, expiration-driven peer activity lagged one quarter, from the pooled response specification without a network interaction. The sample excludes firm–county–quarters in which the responding firm has a same-county lease expiration. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Intensive outcomes condition on positive activity in the corresponding outcome. Standard errors are clustered by formation-block and shown in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. In Panel B, the four signal samples require the listed signal to be positive and statistically distinguishable from zero. These samples overlap and their totals are not additive. The no-measurable-signal sample requires that none of the four signals be statistically distinguishable from zero in either direction. Adoption and investment rates are percentages. Peer activity is the mean of $\widehat{E}^{-i}_{c,q-1}$. All rows use the main sample restriction that excludes firm–county–quarters in which the responding firm has a same-county lease expiration. County–formation years counts the distinct county, formation and calendar-year cells in each sample. Panel A has 285 fewer observations than Panel B because the regressions drop singleton observations absorbed by the fixed effects.

Table OA.22: Vintage-block responses by prior horizontal experience

	Never drilled horizontally	Drilled horizontally elsewhere, not here	Already drilling in this market
Panel A. Technique intensity, extensive adoption			
Early vintage, first block	-0.1889^{**}	0.0010	-3.0723^{**}
	(0.0791)	(0.0590)	(1.2938)
Early vintage, later blocks, ≤2014	0.2285^{***}	0.0122	-1.7792^{***}
	(0.0167)	(0.0500)	(0.6440)
Early vintage, later blocks, 2015+	−0.0195	0.0283	−0.0951
	(0.0185)	(0.0224)	(0.4197)
Late vintage, first block	−0.2964	0.3092	−0.5308
	(0.2311)	(0.2224)	(3.3129)
Late vintage, later blocks, 2017+	−0.0027	0.0543	−1.7379
	(0.0241)	(0.0348)	(2.9478)
Panel B. Peer performance, extensive adoption			
Early vintage, first block	-0.0735^{*}	0.0351	−1.0028
	(0.0399)	(0.0577)	(0.7448)
Early vintage, later blocks, ≤2014	0.1030^{*}	−0.0599	0.2051
	(0.0617)	(0.0722)	(0.6956)
Early vintage, later blocks, 2015+	−0.0008	0.0526	−1.4695
	(0.0096)	(0.0622)	(1.8410)
Late vintage, first block	0.0060	0.1123^{***}	2.1445^{*}
	(0.0376)	(0.0374)	(1.1289)
Late vintage, later blocks, 2017+	0.3369^{***}	0.1104^{*}	−0.6974
	(0.1244)	(0.0593)	(3.3822)
Panel C. Intensive investment, firms already drilling in this market			
	Technique intensity	Peer performance	
Early vintage, first block	17.43	33.04	
	(33.42)	(22.16)	
Early vintage, later blocks, ≤2014	36.85^{***}	37.11^{**}	
	(12.42)	(18.33)	
Early vintage, later blocks, 2015+	-40.92^{**}	10.37	
	(20.72)	(50.62)	
Late vintage, first block	159.63	52.59^{***}	
	(132.87)	(15.36)	
Late vintage, later blocks, 2017+	17.92	117.02^{*}	
	(70.71)	(68.54)	

Notes. The unit is a firm–county–formation–quarter. Each coefficient is on one-quarter-lagged expiration-driven peer activity interacted with the network signal, estimated separately within each vintage-block group and firm group. Firms are divided according to their horizontal drilling before the response quarter. The groups are firms that have never drilled horizontally, firms that have drilled horizontally elsewhere but not in the current county and formation, and firms that have already drilled horizontally in the current county and formation. The vintage-block groups follow main Table 3, except that the early-vintage later-block 2015+ row combines the finer calendar groups of Table 3 and is estimated as one group. Intensive investment can be estimated only for firms already drilling in the market, because the other groups have fewer than 500 observations in each group. Controls, fixed effects and the signal-specific samples follow the main specification. The sample excludes firm–county–quarters in which the firm has an own lease expiring in the same county and quarter. Every regression includes the lower-order peer-activity and signal terms. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive investment is the number of horizontal wells among observations with positive horizontal investment. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.23: Peer activity from the first well on each lease

Panel A. Pooled response

	Extensive adoption		Extensive investment		Intensive investment	
	All wells	First wells	All wells	First wells	All wells	First wells
Peer performance	0.1531**	0.5467**	0.1835**	0.6653**	5.62	48.23***
	(0.0649)	(0.2618)	(0.0756)	(0.2709)	(6.70)	(15.89)
Peer-technique breadth	−0.0020***	−0.0037**	−0.0025***	−0.0044**	−0.3267*	−0.9592**
	(0.0006)	(0.0016)	(0.0007)	(0.0019)	(0.1741)	(0.3953)
Technique intensity	0.2661***	0.4724***	0.2988**	0.4541*	37.61	64.10
	(0.0955)	(0.1749)	(0.1249)	(0.2481)	(27.15)	(39.80)

Panel B. Extensive adoption by period

	2008–14		2015–17		2018 onward	
	All wells	First wells	All wells	First wells	All wells	First wells
Peer performance	−0.0458	−0.0685	0.2822***	0.9641***	−0.3274*	0.6760***
	(0.0573)	(0.1285)	(0.0780)	(0.1568)	(0.1938)	(0.2484)
Technique intensity	−0.1736***	−0.2351	0.3572*	0.4950*	0.2802*	−0.2100
	(0.0354)	(0.1495)	(0.2107)	(0.2995)	(0.1447)	(0.5580)

Panel C. First-block vintage contrast, late minus early, extensive adoption

	All wells	First wells
Technique intensity	1.4216*	2.3189
	(0.7743)	(1.8117)
Peer performance	0.3197**	0.8428***
	(0.1242)	(0.2693)

Notes. "All wells" is the main county-level peer-activity measure. "First wells" keeps only the first well on each lease, ranking the wells linked to a lease by spud date. It keeps the event-time bunching weights unchanged and removes 58 percent of bunching-weighted peer activity. Both versions use the same-county own-expiration exclusion, the full control set and operator, county, operator-by-county and quarter fixed effects. Every response regression includes the lower-order peer-activity and signal terms and uses the corresponding signal-specific sample. Extensive adoption indicates a positive change in the firm's horizontal share. Extensive investment indicates positive horizontal drilling. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Coefficient magnitudes are larger under the first-well measure partly because the measure is smaller in scale. Panel C estimates the first-block late-minus-early contrast from a regression on first-block observations only, with one set of fixed effects and controls shared by both vintages. Table 3 of the main paper instead reports the contrast from the saturated specification with cell-specific fixed effects, so its baseline values (1.286 and 0.378) differ from the "All wells" column here. Standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Peer skill is omitted because its signal-specific sample contains only four formation-blocks, too few for clustered inference.

Table OA.24: Peer activity dated at completion and disclosure timing

Panel A. Pooled response, peer activity dated at spud or at completion

	Extensive adoption		Extensive investment		Intensive investment	
	Spud	Completion	Spud	Completion	Spud	Completion
Peer performance	0.1531**	0.1745	0.1835**	0.2133	5.62	2.68
	(0.0649)	(0.1214)	(0.0756)	(0.1376)	(6.70)	(11.65)
Peer-technique breadth	−0.0020***	−0.0020***	−0.0025***	−0.0023***	−0.3267*	−0.1367
	(0.0006)	(0.0005)	(0.0007)	(0.0005)	(0.1741)	(0.1166)
Technique intensity	0.2661***	0.1166	0.2988**	0.0910	37.61	8.13
	(0.0955)	(0.0838)	(0.1249)	(0.0900)	(27.15)	(11.17)

Table OA.24 (continued): Peer activity dated at completion and disclosure timing

Panel B. Responder wells completed after peers' attributes would be public

	Wells	After first peer +60 days	After median peer +60 days	Median slack days
All responders	15,213	0.83	0.71	67
Technique-intensity sample	9,312	0.81	0.73	82
Peer-performance sample	11,348	0.82	0.70	73
Breadth sample	7,391	0.83	0.72	69
Technique intensity, above median	5,300	0.80	0.72	86
Technique intensity, at or below median	4,012	0.83	0.73	77

Panel C. Drilling speed and crowding, horizontal wells

	Wells	Median spud-to-completion days	Next spud by another operator within 90 days	Operators per county–formation–quarter, mean
All county–formation–quarters	104,281	153	0.93	3.47
Technique-intensity sample	77,733	158	0.93	3.38
Peer-performance sample	64,349	150	0.93	3.48
Breadth sample	59,758	162	0.93	3.59
Technique intensity, above median	50,662	171	0.94	3.50
Technique intensity, at or below median	27,071	137	0.93	3.21

Notes. Panel A reports the pooled response with county-level peer activity dated at the peer well's spud, as in the main text, or at its completion, the spud date plus reported spud-to-completion days. Wells without a completion date are dropped from the completion-dated measure. They carry 8 percent of the bunching-weighted peer activity. Both columns use the full control set, the same fixed effects, the lower-order peer-activity and signal terms, the signal-specific samples, and the same-county own-expiration exclusion. Panel B covers horizontal wells of firms that drill horizontally in a quarter after expiration-driven peer activity in the previous quarter. It reports the share of those wells completed after the first and after the median peer well's completion date plus 60 days, and the median number of days between the median peer's completion plus 60 days and the responder's completion. Panel C covers horizontal wells in the county-formation-quarters of each signal's sample. It reports the median time from spud to completion, the share of wells followed within 90 days by a horizontal spud of another operator in the same county and formation, and the mean number of operators spudding horizontal wells per county-formation-quarter. Signal samples overlap. Technique intensity is split at its median in the technique-intensity sample. Panel A standard errors clustered by formation-block are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.25: Parameter values used in the counterfactual exercise

Parameter	What it governs	Value
m_0	Base return value of local evidence	2.73
m_1	Change in that return value with industry maturity	−1.29
b	Saturation as local development accumulates	1.44
f	Cost of starting in a young industry	8.62
d	Decline in that cost as local development accumulates	1.51

Notes. The table reports the rounded parameter values used in the counterfactual exercise. The values minimize the inverse-variance-weighted sum of squared deviations between the model responses and ten technique-intensity estimates reported in Appendix Table E.1 of the main paper. The targets are extensive adoption and horizontal well counts conditional on positive horizontal investment for five vintage-block groups. Investment estimates and standard errors are multiplied by 0.018588 to express both outcomes on a common scale. Local development is normalized to $n = 0$ for first blocks and $n = 1$ for later blocks. Industry maturity N is the industry-wide horizontal share averaged over each target's estimation sample. The values are illustrative, not structural estimates. Appendix E.3 of the main paper reports the calibration procedure and fitted responses.

Table OA.26: Peer-performance targets

Target	n_k	N_k	Adoption
Early vintage, first block	0	0.741	-0.1440^{*}
			(0.0776)
Early vintage, later blocks through 2014	1	0.564	-0.0687
			(0.0656)
Early vintage, later blocks from 2015	1	0.896	0.3433
			(0.2971)
Late vintage, first block	0	0.820	0.2342^{**}
			(0.0947)
Late vintage, later blocks from 2017	1	0.919	-0.6771^{*}
			(0.3563)

Notes. The five targets are peer-performance extensive-adoption responses for the listed formation-vintage, local-block, and calendar-period groups. Local development is normalized to $n_k = 0$ for first blocks and $n_k = 1$ for later blocks. Industry maturity N_k is the industry-wide horizontal share averaged over target k's adoption estimation sample. Standard errors clustered by formation-block are in parentheses. Standard errors are those used in the calibration, from the per-vintage regressions, and differ slightly from Table 3 of the main paper, which estimates all cells in one pooled regression. Stars denote significance at the 10, 5, and 1 percent levels.

Table OA.27: Intensive-investment responses by formation vintage and block position

Era and local-development state	Early vintage	Late vintage	Late minus early
Panel A. Peer performance, intensive investment			
First-block state	-7.311	38.002^{**}	45.313^{**}
	(6.287)	(18.768)	(19.793)
Later blocks, through 2014	35.407^{**}	—	—
	(16.573)		
Later blocks, 2015–16	-124.739	—	—
	(312.353)		
Later blocks, 2017+	102.945^{***}	114.318^{*}	11.373
	(27.143)	(60.670)	(66.465)
Later blocks, 2019+	137.720^{***}	94.906	-42.814
	(11.899)	(162.785)	(163.219)
Panel B. Technique intensity, intensive investment			
First-block state	59.637	45.086	-14.551
	(73.779)	(157.661)	(174.070)
Later blocks, through 2014	34.407^{***}	—	—
	(11.437)		
Later blocks, 2015–16	-45.523	—	—
	(43.505)		
Later blocks, 2017+	-83.595^{***}	2.439	86.034
	(13.022)	(68.734)	(69.957)
Later blocks, 2019+	85.961	203.225^{***}	117.264
	(59.247)	(68.877)	(90.853)

Notes. The unit is a firm–county–formation–quarter. The table is the intensive-investment counterpart to the main paper's aligned formation-vintage table. Later blocks are divided into observations through 2014, 2015–16 and 2017 onward. The final row re-estimates the common 2019–21 window. Dashes mark periods in which late-vintage formations have no later-block observations. The last column reports directly estimated late-minus-early differences. Regressions include the lower-order terms, baseline controls and fixed effects, use the corresponding positive-signal sample and condition on positive horizontal investment. Coefficients are in horizontal wells per estimated excess peer well per unit of the signal. Standard errors cluster by formation-block. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.28: Pooled responses to expiration-driven peer activity by well-to-lease match radius

Match radius	Extensive adoption	Intensive investment
Panel A. Peer performance [distance-time]		
0.10 mi.	0.4383***	26.87
	(0.1253)	(18.71)
0.25 mi. (baseline)	0.1531**	5.62
	(0.0649)	(6.70)
0.50 mi.	0.0833**	1.86
	(0.0371)	(3.34)
1.00 mi.	0.0510**	1.06
	(0.0242)	(2.11)
2.00 mi.	0.0340*	1.22
	(0.0206)	(1.90)
Full county	0.0299	1.14
	(0.0190)	(1.84)
Panel B. Peer skill [distance-time]		
0.10 mi.	0.3456***	−19.83
	(0.0501)	(26.38)
0.25 mi. (baseline)	0.0404	−37.46**
	(0.1688)	(15.57)
0.50 mi.	−0.0069	−1.73
	(0.0136)	(4.65)
1.00 mi.	0.0003	1.86
	(0.0099)	(1.53)
2.00 mi.	−0.0013	1.75
	(0.0102)	(1.57)
Full county	−0.0017	1.75
	(0.0102)	(1.57)
Panel C. Peer-technique breadth [equidistant]		
0.10 mi.	−0.0059*	−0.8266***
	(0.0031)	(0.1891)
0.25 mi. (baseline)	−0.0020***	−0.3267*
	(0.0006)	(0.1741)
0.50 mi.	−0.0007***	−0.0389
	(0.0003)	(0.0340)
1.00 mi.	−0.0005***	−0.0129
	(0.0002)	(0.0182)
2.00 mi.	−0.0005***	−0.0104
	(0.0002)	(0.0161)
Full county	−0.0005***	−0.0084
	(0.0002)	(0.0156)
Panel D. Technique intensity [distance-time]		
0.10 mi.	0.8643***	97.98**
	(0.3225)	(44.49)
0.25 mi. (baseline)	0.2661***	37.61
	(0.0955)	(27.15)
0.50 mi.	0.1054	8.57
	(0.0777)	(6.23)
1.00 mi.	0.0542	4.36
	(0.0449)	(3.15)
2.00 mi.	0.0452	4.11
	(0.0357)	(2.55)
Full county	0.0440	3.80
	(0.0333)	(2.43)

Notes. The unit is a firm–county–formation–quarter. Each panel reports the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed network signal. Rows vary the maximum well-to-lease match radius used to construct peer activity. The 0.25-mile radius is the baseline. Standard errors are in parentheses. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Signal-specific samples require the listed signal to be positive and statistically different from zero. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. Within each signal and outcome the sample size is the same at every radius. Extensive adoption uses 582,531 observations for peer performance, 152,297 for peer skill, 683,440 for peer-technique breadth and 778,002 for technique intensity. Intensive investment uses 8,565, 1,080, 8,420 and 10,348 observations in the same order. Standard errors are clustered by formation-block. Web Appendix Table W.2 reports all four outcomes. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.